MATHEMATICS GRADE 10: STATISTICS I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.1 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 4.0: Statistics and Probability
Sub-Strand	Sub-Strand 4.1: Statistics I
Total Duration	9 lessons (approximately 18 periods × 40 minutes = 720 minutes)
Sub-Strand Content	– Frequency distribution tables for ungrouped data – Simple distribution tables and tally charts – Measures of central tendency: mean of ungrouped data – Measures of central tendency: mode of ungrouped data – Measures of central tendency: median of ungrouped data – Representation of data: line graphs – Representation of data: bar graphs – Representation of data: pie charts (construction) – Interpretation of graphs and pie charts – Importance of statistics in real-life decision-making
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) Collect, organize, and represent ungrouped data in frequency distribution tables b) Calculate the mean, mode, and median of ungrouped data sets drawn from real-life Kenyan contexts c) Represent data accurately using line graphs, bar graphs, and pie charts d) Read, interpret, and draw conclusions from graphs and pie charts e) Appreciate the importance of collecting, organizing, and interpreting data in making effective decisions in everyday life
Core Competencies	– Communication and Collaboration: Learners discuss and compare their frequency tables, graphs, and measures of central tendency with peers, articulating mathematical reasoning clearly during group activities – Critical Thinking and Problem Solving: Learners analyze ungrouped data sets (e.g., maize harvest yields across Rift Valley counties), identify patterns, calculate central tendency measures, and draw evidence-based conclusions – Digital Literacy: Learners engage with PhET simulations and online videos to explore mean, mode, and median interactively, building confidence with digital learning tools – Learning to Learn: Learners reflect on errors in graph construction and interpretation, revising their models and building metacognitive awareness of statistical reasoning – Self-Efficacy: Learners build confidence by successfully constructing and interpreting their own pie charts and bar graphs from locally sourced Kenyan data

Core Values	– Responsibility: Learners appreciate the responsibility of accurate data collection and honest representation, understanding that misleading statistics can harm communities (e.g., misreporting crop yields or health data) – Integrity: Learners are guided to record and present data truthfully, even when results are unexpected, reinforcing honest mathematical practice – Social Justice: Learners examine how statistical data on issues like school attendance, food distribution, or water access in counties such as Turkana or Kisumu can drive equitable policy decisions – Unity: Collaborative data-collection activities foster teamwork and mutual respect among learners from diverse Kenyan backgrounds – Patriotism: Learners use real Kenyan data sets (KNBS census figures, KCPE scores, rainfall data) connecting mathematics to national development and civic awareness
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners formulate questions about patterns in Kenyan data sets (e.g., "What is the most common number of siblings in our class?") that drive the need for frequency distribution tables – Analyzing and Interpreting Data: Central to every lesson — learners organize raw data into tables, compute measures of central tendency, and construct graphs to reveal trends – Using Mathematics and Computational Thinking: Learners apply arithmetic operations to calculate mean, determine mode by frequency inspection, and locate median through ordered data sets – Constructing Explanations: Learners write and present explanations of what their graphs and central tendency values reveal about real-world Kenyan phenomena (e.g., daily mandazi sales at a Nairobi market) – Obtaining, Evaluating, and Communicating Information: Learners critically evaluate graphs encountered in Kenyan newspapers and government reports, identifying accurate versus misleading representations
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: Learners apply statistics to interpret market price data, household income distributions, and budgeting scenarios relevant to Kenyan families – Health Education: Learners use data on nutrition (e.g., protein content in githeri versus nyama choma) and disease prevalence to see how statistics informs public health decisions in Kenya – Environmental Education: Learners analyse rainfall and temperature data from Kenyan weather stations (e.g., Kericho tea-growing region vs. arid Turkana) to understand climate variability through statistical lenses – Life Skills: Learners develop decision-making and critical thinking skills by interpreting data before drawing conclusions, a skill transferable to everyday Kenyan life contexts – Citizenship: Learners examine how the Kenya National Bureau of Statistics (KNBS) uses data collection and representation to inform national planning, budgets, and the census
Career Connections	– Data Analyst / Statistician: This sub-strand lays the foundation for careers at institutions like the Kenya National Bureau of Statistics (KNBS) or private research firms, where organizing and interpreting large data sets is core work – Medical / Public Health Officer: Epidemiologists and public health workers in Kenya (e.g., at KEMRI) rely on measures of central tendency and graphical data to track disease outbreaks and design interventions – Agricultural Extension Officer: Officers advising Kenyan smallholder farmers use statistical tools to analyse crop yield data, advise on planting decisions, and report to county governments – Business / Economics Analyst: Market analysts at Nairobi Stock Exchange or retail chains use frequency distributions and graphs to understand consumer trends and sales performance – Journalist / Media Professional: Data journalists at outlets like Nation Media Group use graphs and statistics to communicate findings on elections, health, and economy to the public – Teacher / Educator: Mathematics and science teachers require deep understanding of statistics to teach CBC effectively across Kenyan schools

Focus for Lessons	This unit introduces Grade 10 learners to the foundational concepts of Statistics through the lens of real Kenyan data — from KCPE score distributions in Nairobi schools to maize harvest records in the Rift Valley. Learners progress from organizing raw data into frequency distribution tables, through calculating measures of central tendency (mean, mode, and median), to constructing and interpreting line graphs, bar graphs, and pie charts. The unit emphasizes that statistics is not merely a computational exercise but a powerful tool for making sense of the world — enabling communities, governments, and individuals across Kenya to make informed, evidence-based decisions.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action? KICD KEY INQUIRY QUESTIONS: 1. How do we collect, organize, and represent data? 2. How do measures of central tendency help us make sense of data in real life?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Maize Harvest Mystery of Nakuru County" A video is shown of a Kenya Agricultural and Livestock Research Organization (KALRO) officer visiting five smallholder maize farmers in Nakuru County. Each farmer reports their total maize harvest in 90-kg bags: the values are scattered and seemingly random — some farmers harvested as few as 4 bags while others harvested as many as 31 bags. The officer writes all 20 farmers' values on a chalkboard in the order she collected them. The data looks chaotic and impossible to interpret at a glance. The puzzle: The county government needs ONE number to represent the "typical" harvest for Nakuru County farmers so they can decide how much subsidized fertilizer to allocate next season. But looking at the raw list, students notice that different farmers suggest different "representative" numbers — one says use the most common value, another says use the middle value, another says add them all up and divide. The video ends with the officer asking: "Which single number best represents all our farmers — and how do we even begin to make sense of all this data?" This is puzzling because learners immediately feel the tension: raw data is overwhelming and unhelpful until it is organized. They are motivated to find tools (frequency tables, averages, graphs) that turn a chaotic list into a story that can drive real decisions affecting Kenyan farmers' livelihoods.
Storyline Thread	Lesson 1 — "Making Sense of Chaos" (Frequency Distribution Tables): Students encounter the anchoring phenomenon (Nakuru maize harvest data). They attempt to make sense of 20 raw data values, discover the need for organization, and construct their first frequency distribution table for ungrouped data using tally marks. They add their first "wonder" question to the Driving Question Board (DQB). Lesson 2 — "Sorting the Story" (Simple Distribution Tables & Tally Charts): Students refine their frequency tables using a video on simple distribution tables, practice with a new data set (daily sukuma wiki prices at Kongowea Market, Mombasa), and compare how a well-structured table makes patterns immediately visible. DQB is updated with new observations. Lesson 3 — "Finding the Balance Point" (Mean of Ungrouped Data): Using the PhET simulation, students explore the concept of mean as a balance point. They calculate the mean of the Nakuru maize harvest data and the mandazi sales data, connecting the computed value back to the anchoring phenomenon — does the mean represent the "typical" farmer?

	Lesson 4 — "The Most Popular Value" (Mode of Ungrouped Data): Students investigate mode through the mandazi market supporting phenomenon, identifying the most frequently occurring value. They conduct a classroom data-collection activity (e.g., number of siblings), find the mode, and debate whether the mode or mean is more useful for the fertilizer allocation decision. Lesson 5 — "The Middle of the Story" (Median of Ungrouped Data): Students order data sets and locate the median, using the KCPE Kisumu supporting phenomenon to see why median can be more representative than mean when extreme values exist. They compare mean, mode, and median for the Nakuru data set, updating their DQB with a refined answer to the driving question. Lesson 6 — "Drawing the Journey" (Line Graphs): Using the Kericho rainfall supporting phenomenon and a video on line graphs, students construct line graphs from provided data sets (e.g., monthly maize prices in Nairobi). They identify trends, peaks, and troughs, connecting graphical patterns to real agricultural and economic decisions. Lesson 7 — "Comparing with Bars" (Bar Graphs): Students watch a video on bar graphs and construct their own bar graphs from frequency distribution tables built in Lessons 1–2. They compare multiple categories (e.g., harvest yields across five Rift Valley counties) and discuss how bar graphs communicate comparisons more powerfully than raw tables. Lesson 8 — "Slicing the Pie" (Construction of Pie Charts): Using the Kenya Energy Mix supporting phenomenon as a model, students learn to calculate angles from percentages and construct accurate pie charts by hand. They create pie charts representing the class data collected in Lesson 4, connecting construction skills to real-world data visualization. Lesson 9 — "Reading the Full Picture" (Interpretation of Graphs, Pie Charts & Importance of Statistics): Students interpret a set of mixed graphs and pie charts drawn from Kenyan newspaper articles and KNBS reports. They write a short advisory report to the fictional Nakuru County government recommending a fertilizer allocation strategy, using all statistical tools learned in the unit. The lesson closes with a video on the importance of statistics in real life and a full DQB review — students answer the driving question using evidence gathered across all 9 lessons.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "Mandazi Sales at Gikomba Market": A Nairobi market vendor records the number of mandazi sold each day for 20 days. The raw tally is shown on screen — students see how a simple frequency distribution table instantly reveals the most common daily sales figure (mode), helping the vendor decide how many to bake each morning. – SUPPORTING PHENOMENON 2 — "Rainfall Records in Kericho Tea Country": A line graph from the Kenya Meteorological Department showing monthly rainfall in Kericho over one year is displayed alongside raw monthly figures. Students observe how the line graph reveals a wet-season peak invisible in the raw numbers, motivating the power of graphical data representation. – SUPPORTING PHENOMENON 3 — "KCPE Score Distribution in a Kisumu School": A bar graph of KCPE scores from a Kisumu primary school is shown with the mean score marked. Students notice several pupils scored far above and below the mean, prompting discussion of whether the mean alone tells the full story — introducing the value of median and mode alongside mean. – SUPPORTING PHENOMENON 4 — "Kenya's Energy Mix Pie Chart": A pie chart from the Kenya Energy and Petroleum Regulatory Authority (EPRA) showing Kenya's electricity generation sources (geothermal, hydro, wind, solar, thermal) is displayed. Students are challenged to reverse-engineer the chart — given the percentages, can they reconstruct it? This motivates the lesson on pie chart construction.

LESSON 1 (80 minutes): Making Sense of Chaos: Frequency Distribution Tables	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce students to the anchoring phenomenon and motivate the need for organizing raw data using frequency distribution tables for ungrouped data.
Knowledge	Students will know that a frequency distribution table organizes raw data by listing each value (or category) alongside the number of times it occurs (its frequency), and that tally marks are a systematic tool for counting occurrences within a dataset.
Skills	Students will be able to collect raw data values, construct a frequency distribution table for ungrouped data using tally marks, and read frequencies from a completed table.
Attitudes	Students will appreciate the value of organized data as a tool for informed decision-making in their communities, and develop curiosity about how mathematics helps solve real-world Kenyan agricultural challenges.
Key Inquiry Question	How do we collect, organize, and represent data?
Purpose in Storyline	This is the opening lesson of the unit. Students encounter the anchoring phenomenon — the Maize Harvest Mystery of Nakuru County — and immediately feel the tension of trying to make sense of 20 raw, unorganized data values. By constructing their first frequency distribution table, students discover that organizing data reveals patterns invisible in raw lists. This lesson launches the Driving Question Board and sets the investigative thread that the entire unit will follow.
Safety Notes	No physical safety concerns for this lesson. Ensure the video viewing environment is respectful — remind students that the farmers depicted represent real livelihoods and should be discussed with dignity and empathy.

B. LESSON OVERVIEW
Students watch a short video depicting a KALRO officer collecting maize harvest data from 20 smallholder farmers in Nakuru County. The raw data — 20 values listed in collection order on a chalkboard — is chaotic and difficult to interpret. Students first predict what the "typical" harvest might be just by scanning the raw list, then observe the data more carefully through a structured activity. They discover the limitations of raw data and, guided by the teacher, construct a frequency distribution table using tally marks to organize the values. The lesson closes with students posting their first "wonder" questions on the Driving Question Board, setting up the investigative arc for the entire Statistics I unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Regrouping whole number place values] Source: Khan	Students are shown the 20 raw maize harvest values (in 90-kg bags) from Nakuru County farmers, displayed on the board exactly as the KALRO officer wrote them — in collection order,	Write the 20 raw values on the board in a single unorganized list before students enter, exactly as follows: 12, 7, 19, 4, 31, 12, 8, 15, 12, 22, 7, 18, 31, 5, 12, 9, 15, 7, 20, 12. Say: 'A KALRO officer	Strategy: Activate-and-Confront Prediction. Students commit to a prediction before instruction so that the subsequent lesson creates productive cognitive conflict. When students later discover their	Observable indicator: Every student has written a numerical prediction and at least one question in their notebook within the 2-minute window. Teacher circulates and notes (a) the range

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	unorganized. Without any instruction, each student silently writes in their notebook: (a) their best guess at the 'typical' harvest for a Nakuru farmer, (b) one word describing how looking at the raw list makes them feel (e.g., confused, curious, overwhelmed), and (c) one question the data makes them want to ask. Students then share predictions with a shoulder partner before three students share with the whole class.	visited 20 maize farmers in Nakuru County last harvest season. She wrote down exactly what each farmer told her — this is the real data, in the order she collected it. Each number is how many 90-kg bags of maize that farmer harvested. Before we watch the video, look at this list for 30 seconds — do not calculate anything. Then answer the three questions in your notebook.' Allow 2 minutes of silent individual writing. Then say: 'Turn to your shoulder partner and share your typical harvest guess and one question. You have 90 seconds.' After partner talk, cold-call exactly 3 students: one who guessed a low value, one who guessed a high value, one who guessed a middle value (scan notebooks as students write to identify these). Ask: 'Why did you choose that number? What were you looking at?' Apply 10–15 seconds of wait time after each response before probing further. Record all three guesses on a corner of the board labelled 'Our Predictions — We Will Return to These.'	intuitive guesses were inconsistent with each other — and that different methods yield different 'typical' values — they are motivated to find a principled way to resolve the disagreement. This anchors all subsequent learning in a felt need rather than abstract instruction.	of predictions made — a wide spread indicates students are genuinely uncertain, which is the desired starting state; (b) whether students' questions mention fairness, accuracy, or method (higher-order) vs. simple arithmetic (surface-level). Teacher uses this to calibrate how much scaffolding to provide in the Explain phase. Do NOT correct any predictions at this stage.
Observe Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12	Students watch a 4–5 minute teacher-narrated video (or teacher-led roleplay if video is unavailable) depicting the KALRO officer visiting five of the twenty farmers in Nakuru County. The officer's chalkboard fills up with values as she goes. At the video's end, the officer poses the central puzzle: 'The county government needs ONE number to represent the typical harvest — which number should I give them, and how do I even begin to make sense of all this data?' After the video, students complete a structured	Before playing the video say: 'As you watch, listen carefully for the three things different farmers suggest when asked what the typical harvest is. You will need those three suggestions in a moment.' Play the video (or enact the roleplay). Pause at the moment the officer writes all 20 values on her chalkboard and say: 'Freeze your thinking here. Look at this list. The county government officer has exactly 10 minutes before her next meeting. She needs one number. What is she feeling right now?' Allow 10	Strategy: Notice and Wonder (Inquiry Anchoring). The structured Notice/Wonder protocol ensures all learners — including those who struggle with immediate calculation — have a valid entry point into the data. Noticing observable features (e.g., '12 appears several times') precedes and scaffolds the later cognitive work of organizing. The deliberate failed attempt to find the mode by eye creates authentic motivation for the tally-and-table method in the Explain phase.	Observable indicator: Students' 'I Wonder' entries reveal their intuitive proximity to key statistical concepts. Look for wonders such as: 'Why are some farmers harvesting so much more than others?' (variability), 'What number is in the middle?' (median), 'What if we added them all up?' (mean). Teacher mentally notes which concepts are already surfacing — these will be used to sequence the DQB in Phase 4. The failed counting attempt is itself formative: if more than half the class gets inconsistent answers, this confirms

🔍 Search ARES for similar readings	'Notice and Wonder' table in their notebooks with two columns — 'I Notice' and 'I Wonder' — recording at least two entries in each column. Students then attempt individually (5 minutes, no guidance) to answer: 'Which farmer's harvest value appears most often in the list?' using only their eyes and the raw list on the board.	seconds of wait time. After the video ends, say: 'In your notebook, open a fresh page and draw a two-column table: left column — I Notice, right column — I Wonder. You have 3 minutes to fill it in silently.' Circulate, reading over shoulders. After 3 minutes, say: 'Now, without talking to anyone, try to find which harvest value appears most often. Circle it on your own copy of the data. You have 5 minutes.' After the attempt, cold-call 4 students and ask them to state their answer and their method. Expect variation — some will recount multiple times, some will lose track, some will get different answers. Do NOT resolve the disagreement yet. Say: 'Interesting — we have different answers. Why might that be? Hold that thought.'		the need for frequency tables and can be named explicitly in the Explain phase.
Explain Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the concept of a frequency distribution table through a guided class construction. Using the 20 Nakuru harvest values, the whole class builds the table together on the board — first listing all unique values in ascending order, then using tally marks to count occurrences, then recording the final frequency for each value. Students copy the table into their notebooks and answer three interpretation questions: (1) Which harvest value is most common among Nakuru farmers? (2) How many farmers harvested fewer than 10 bags? (3) What does the frequency column tell us that the raw list could not? Students then work in pairs to construct a second frequency distribution table from a new small dataset: the number of mandazi sold daily by 15 vendors	Begin by returning to the failed counting attempts: 'A moment ago, four of you gave me four different answers for the most common value. That is not a mistake — it is what raw data does to human brains. Mathematicians faced this same problem and invented a tool. Let me show you.' Draw a blank three-column table on the board with headers: 'Harvest (bags) | Tally | Frequency.' Say: 'We are going to build this table together. My job is to organize — your job is to count and call out.' Point to each of the 20 values in the raw list one at a time. For the value 4, say: 'Four bags — who can find every time 4 appears in the list? Raise your hand when you see one.' Call out tally marks as students identify occurrences. Continue for all unique values (4, 5, 7, 8, 9, 12, 15, 18, 19, 20, 22, 31). After	Strategy: Collaborative Guided Construction (Gradual Release). The teacher and students build the first frequency table together, ensuring every student sees and participates in the process before attempting one independently. The second dataset (mandazi sales) maintains the Kenya context while offering a lower-stakes practice environment. Asking students to point to evidence in the table — rather than just state answers — builds the habit of grounding claims in organized data, which is foundational for all subsequent lessons on measures of central tendency.	Observable indicators: (1) During the joint table construction, teacher checks that all students' tally marks group correctly in sets of five — a common error is failing to cross the fifth tally. (2) For the three interpretation questions, correct answers are: Q1 = 12 bags (frequency 5, the mode), Q2 = 4 farmers (values 4, 5, 7, 7, 7, 8, 9 — teacher must confirm students are reading the frequency column accurately), Q3 = open-ended but should mention 'patterns,' 'easier to see,' or 'how many times.' (3) For the pair activity, both partners' tables must show identical tally and frequency columns — discrepancies signal miscounting and should be addressed before the next lesson.

	at Nakuru's Maasai Market over one week.	completing the table ask: 'What do you notice now that you could not see before?' Wait 12 seconds. Cold-call 3 students. Then say: 'Answer the three interpretation questions in your notebook — 4 minutes, silent work.' After 4 minutes, cold-call one student per question, asking them to point to exactly where in the table they found the evidence. For the pair activity, distribute the mandazi dataset (or write on board): 8, 12, 8, 15, 10, 12, 8, 10, 12, 8, 15, 10, 12, 8, 10. Say: 'With your partner, construct a full frequency distribution table for this data. You have 7 minutes. Use tally marks — no shortcuts.' Circulate and check that tally marks are being grouped in fives correctly. After 7 minutes, one pair presents their table to the class. Ask the class: 'Does everyone's table match? If not, where did the difference come from?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student writes one genuine 'wonder' question sparked by today's lesson on a sticky note (or folded paper card) and posts it on the class Driving Question Board. The teacher reads all questions aloud and clusters them into emerging categories on the board. Students vote (by placing a dot sticker or pencil mark) on the two questions they most want the unit to answer. The board remains displayed throughout the unit and will be revisited at the end of every lesson.	Say: 'The KALRO officer now has a beautiful frequency table. But look — she still doesn't have ONE number to give the county government. She has a table full of numbers. So we have new questions. On your card, write the ONE question you most want mathematics to answer about the Nakuru harvest data — or about data in general. Be specific. Not just "why" — write a question a mathematician could actually investigate.' Give students 2 minutes to write. Collect all cards and read each one aloud as you post it on the DQB. Group similar questions into clusters without naming the clusters yet — let students suggest category names. Likely clusters will emerge around:	Strategy: Driving Question Board (DQB) as a Living Knowledge Map. The DQB externalizes the class's collective curiosity and makes the learning journey visible. Returning to it at the end of every lesson allows students to see their own growing understanding. Clustering questions previews unit structure without front-loading content, and the voting protocol gives students agency over the learning sequence, increasing motivation and ownership.	Observable indicator: The quality of DQB questions reveals students' current conceptual proximity to the unit's key ideas. High-quality questions for this stage include: 'Which single number is most fair to use — the middle, the most common, or the average?' or 'What if we drew the data as a picture — would it be easier to understand?' Surface-level questions (e.g., 'What is a graph?') indicate students need more grounding in the phenomenon context. Teacher photographs the DQB at the end of Lesson 1 for comparison with later lessons.

		(a) finding a single representative number, (b) showing data visually/graphically, (c) fairness and outliers, (d) comparing different groups. After clustering, say: 'Each of these clusters is a door we are going to walk through in this unit. By the end, you will be able to answer every single one of these questions — and you will help the KALRO officer make her recommendation to Nakuru County government.' Point to the predictions from Phase 1 still on the board: 'We will also come back to these predictions. Were any of you right? We will find out.'		
Model Building Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students create their first individual 'Data Story Map' — a one-page visual model in their notebooks that they will add to across the unit. For Lesson 1, the map contains: (a) a sketch of the Nakuru phenomenon (raw chaotic list → frequency table → ???), (b) the completed frequency distribution table for the maize harvest data, (c) an arrow labelled 'What we still don't know,' pointing to an empty box they will fill in future lessons. Students label what the frequency table can and cannot tell them about the Nakuru farmers' harvest.	Say: 'Every lesson in this unit, you are going to update a Data Story Map — a model that shows how we are making sense of the Nakuru harvest mystery. Today you are building the first version. It will look incomplete — that is intentional. Mathematics builds understanding step by step.' Draw a simple three-box diagram on the board: Box 1 labelled 'Raw Data (Chaos),' an arrow, Box 2 labelled 'Frequency Table (Order),' another arrow, Box 3 labelled '??? (Still needed).' Say: 'Copy this structure, then fill in Box 1 with a quick sketch or summary of the raw list, and Box 2 with your completed frequency table. In the arrow between Box 2 and Box 3, write: What does the table still NOT tell the KALRO officer?' Allow 6 minutes. Cold-call two students to share what they wrote in the arrow. Expected responses: 'It doesn't give one number,' 'It doesn't tell us what the average is,' 'It doesn't show a picture of the data.' Affirm all reasonable responses: 'Exactly — your model	Strategy: Iterative Model Building (Cumulative Sensemaking). The Data Story Map is a running model that students revise across the unit — it mirrors the scientific and mathematical practice of refining models as understanding grows. Beginning the map in Lesson 1 with a deliberate gap ('???') ensures students end the lesson with productive incompleteness: they know more than they did at the start, but they can clearly see and name what they still need to discover. This sustains motivation and connects each lesson to the broader investigative arc.	Observable indicator: In the arrow between Box 2 and Box 3, students should articulate a specific limitation of the frequency table — not just 'we need more information' but naming what type: a single number, a visual representation, a way to compare. Students who write only 'I don't know' need targeted support in Lesson 2 through a worked example before independent practice. Teacher collects 5 randomly selected Data Story Maps at the end of the lesson to review before Lesson 2.

		is telling you what you need to learn next. That is what a good model does.' Instruct students to keep the Data Story Map in their notebooks — they will return to it in Lesson 2.		

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions before planning Lesson 2: (1) What was the range of predictions students made in Phase 1 — did the spread suggest genuine uncertainty about representative values, or did students cluster around one number? (2) Which DQB question clusters received the most votes — does the class lean toward wanting a single summary number (pointing toward mean/median/mode) or toward wanting a visual representation (pointing toward graphs and frequency polygons)? This will help you sequence Lessons 2–4. (3) During the frequency table construction, how many students made tally-grouping errors? If more than one-third of the class miscounted, open Lesson 2 with a 5-minute re-do using a fresh small dataset before moving to grouped data. (4) Did any student's Data Story Map gap-arrow reveal a particularly insightful limitation — e.g., noting that the table treats all values equally even when some farmers' conditions are very different? Surface this student's thinking at the start of Lesson 2 as a bridge to discussing why different measures of central tendency exist.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched a KALRO officer collect maize harvest data from 20 Nakuru County farmers and saw 20 raw values written on a chalkboard in collection order. They observed that different students got different answers when trying to find the most common value by eye, and that the raw list made it nearly impossible to see patterns or answer the county government's question about a typical harvest.
What did I learn?	Students learned that a frequency distribution table organizes raw data by listing each unique value alongside tally marks and a final frequency count, making it possible to see at a glance how often each value occurs. They learned that tally marks are grouped in fives to make counting efficient, and that an organized frequency table reveals patterns — such as which value is most common — that are invisible in a raw unordered list.
How does this explain the phenomenon?	Students explained that the KALRO officer's raw data list was difficult to interpret because it showed no organization or pattern, and that constructing a frequency distribution table transformed that chaos into a structured summary. However, they also explained that the frequency table alone still cannot give the county government the single representative number it needs — pointing forward to the need for measures of central tendency in upcoming lessons.

LESSON 2 (80 minutes): Sorting the Story: Simple Distribution Tables & Tally Charts	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to organise raw data into frequency distribution tables and tally charts so that patterns in community data become visible and usable for decision-making.
Knowledge	Learners will know that a frequency distribution table organises data by listing each value or category alongside the number of times it occurs (its frequency), and that tally marks are a systematic counting tool used to build such tables accurately.
Skills	Learners will be able to collect raw data from a given context, construct a tally chart, complete a frequency distribution table, and read patterns (clustering, spread, most common values) directly from the table.
Attitudes	Learners will appreciate the value of organised data as a tool for fairness and informed community decision-making, developing confidence in handling real-life numerical information.
Key Inquiry Question	How do we collect, organise, and represent data?
Purpose in Storyline	In Lesson 1, students encountered the chaos of the KALRO officer's raw harvest list and felt the need for organisation. In Lesson 2 — 'Sorting the Story' — students learn to transform that chaos into a structured frequency distribution table, using a new Kenya-based dataset (daily sukuma wiki prices at Kongowea Market, Mombasa) to practise. By the end, learners can see patterns hiding in raw numbers and update the Driving Question Board with new evidence that moves the class closer to answering: which single number best represents Nakuru's typical maize harvest?
Safety Notes	No physical safety concerns. Ensure inclusive participation — some learners may have limited experience with markets or farming; validate all contexts as equally relevant. Avoid singling out learners who struggle with arithmetic; use peer-supported table construction throughout.

B. LESSON OVERVIEW

Learners revisit the chaotic raw maize-harvest data from Lesson 1 and articulate why it was hard to interpret. They then watch a short instructional video showing a KALRO data clerk constructing a frequency distribution table and tally chart from a similar dataset. Using a new, relatable dataset — daily sukuma wiki prices (in KES) recorded over 20 market days at Kongowea Market, Mombasa — learners construct their own tally charts and frequency tables in pairs, then analyse the completed tables to identify patterns. A whole-class discussion connects these organisational tools back to the Nakuru Maize Harvest Mystery, and the Driving Question Board is updated with new evidence and emerging questions. Learners close by revising their personal predictive models from Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Regrouping whole number]	Learners are shown the same chaotic raw list of 20 maize harvest values from Lesson 1 (written on the board in the order	Write the 20 raw values on the board exactly as listed — do NOT sort them. Say: 'This is the same list our KALRO officer wrote down	Activation of Prior Knowledge & Prediction: By asking learners to strategise before instruction, the teacher surfaces intuitive	Observable indicator: Learners can articulate at least one logical reason why raw unsorted data is difficult to interpret and propose at

place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the KALRO officer collected them: 4, 31, 12, 9, 17, 4, 22, 8, 31, 14, 9, 4, 17, 25, 12, 9, 17, 4, 22, 14). They are asked: 'Without doing any calculations, if you had to organise this list to make it easier for the county government officer to understand at a glance — what would you do first? Write down your strategy in your exercise book.' Learners work individually for 4 minutes, then share with a partner for 2 minutes.	in Nakuru. The county government needs to understand it quickly. Before I teach you anything new today, I want YOUR brain to work. In your exercise book, write: what is the very first thing you would do to this list to make it easier to read?' Allow 4 minutes of silent individual thinking. Circulate and note 3–4 different strategies learners have written (e.g., sorting smallest to largest, grouping same numbers, drawing a picture). After 4 minutes say: 'Now share your strategy with your partner. You have 2 minutes.' After sharing, cold-call exactly 4 learners — choose deliberate variety: one who sorted, one who grouped, one who drew, one with an unexpected idea. For each, ask: 'Why did you choose that approach?' Allow 10 seconds of WAIT TIME after each response before reacting. Do not confirm or correct yet — record all strategies on the board under the heading 'Our Organising Ideas.'	organising ideas and creates a felt need for a systematic method. Recording all ideas without judgment lowers affective barriers and primes learners to evaluate the instructional video against their own thinking.	least one organising action (sorting, grouping, tallying, graphing). Teacher scans written responses during circulation — flag learners who leave the page blank for targeted support during the video phase.
Observe Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners watch a 7-minute instructional video: 'How a KALRO Data Clerk Organises Harvest Data.' The video shows a clerk in Nakuru taking a raw list of 20 maize harvest values, constructing a tally chart step-by-step (reading each value, finding its row, marking a tally), then converting tally counts into a clean frequency distribution table with columns: Harvest (bags) | Tally | Frequency. The clerk narrates each decision aloud. After the video, learners receive a printed or board-written dataset: 'Daily price of sukuma wiki (KES per bunch) at Kongowea Market, Mombasa over 20 market days: 10, 15, 10, 20, 15, 10, 25, 20, 15, 10, 20, 15, 25, 10, 15, 20,	Before playing the video say: 'As you watch, I want you to do two things: first, tick any organising idea from our board that the clerk also uses; second, notice one thing the clerk does that nobody on our board suggested.' Play the video without pausing unless technology fails. After the video, give learners 30 seconds of silent reflection, then say: 'Which of our ideas did the clerk use?' Cold-call 2 learners. Then ask: 'What did the clerk do that we didn't think of?' Cold-call 2 different learners. Allow 10 seconds WAIT TIME each time. Then say: 'Now it's your turn. Here is data from Kongowea Market in Mombasa — a place where sukuma wiki prices change day to	Worked Example then Immediate Transfer: The video provides a fully modelled worked example in a Kenyan context (KALRO/Nakuru), immediately followed by a parallel task in a new Kenyan context (Kongowea Market/Mombasa). This 'I do, we do' sequence lets learners test comprehension in real time rather than passively absorbing the model. The sukuma wiki context is deliberately chosen as universally familiar across Kenya, reducing cognitive load on context so learners can focus on the mathematical procedure.	Observable indicator: Each pair produces a tally chart with exactly 4 distinct price rows (10, 15, 20, 25 KES) and tally marks that sum to 20. Frequency column totals must also sum to 20. Teacher checks at least 6 pairs during circulation. Common error to watch for: learners who tally in rows of 4 instead of the standard groups of 5 (four vertical, one diagonal). Address immediately by pointing to the video's method and asking: 'Why do we group tallies in fives?'

	10, 15, 20, 25.' In pairs, learners construct a tally chart and frequency distribution table for this dataset in their exercise books.	day, just like maize harvests change farm to farm. With your partner, build a tally chart and then a frequency table. Use the same column headings the clerk used: Price (KES) | Tally | Frequency. You have 15 minutes.' Circulate actively — every 3 minutes, stop and ask one pair to show their tally marks so far on the board. Prompt: 'How do you know your tally marks are correct? What could you do to check?' If a pair finishes early, pose the extension: 'Add a fourth column: Relative Frequency (fraction of 20 days). What does each fraction tell the market vendor?'		
Explain Phase 📹 **VIDEO:** Regrouping whole number place values Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Pairs share their completed frequency tables. The class constructs a single agreed 'master table' on the board. Learners then answer three analysis questions individually in their exercise books: (1) Which sukuma wiki price occurred most often? (2) On how many days was the price above 15 KES? (3) A Kongowea Market vendor wants to stock up on days when prices are lowest — what does the table tell her? Groups then discuss: 'How is our sukuma wiki table similar to what the KALRO officer needed for the maize harvest data? How is it different?' Finally, learners are asked: 'Could you use this same table structure on the Nakuru maize harvest data from Lesson 1? What would the rows be?'	Ask one pair to read out their frequency row for KES 10. Ask the class: 'Does everyone agree? Thumbs up, thumbs sideways, thumbs down.' Resolve any disagreements by re-tallying together aloud. Build the master table on the board column by column. Once complete, say: 'Now use only the table — not the raw data — to answer these three questions in your exercise book. You have 6 minutes. Silent, individual work.' After 6 minutes, cold-call 3 different learners (one per question) — for each, ask 'How did the table help you answer that — could you have answered it from the raw list as quickly?' Allow 10 seconds WAIT TIME. For the discussion question comparing sukuma wiki to maize harvest, give pairs 3 minutes, then cold-call 2 pairs. Push deeper: 'The KALRO officer has values ranging from 4 to 31 bags — how many rows would her table need? Is there a smarter way to handle a wide spread of values?' (Do not answer	Table-to-Insight Bridge: By requiring learners to answer interpretive questions using only the completed table (not the raw data), the teacher forces learners to experience directly how a frequency table compresses information into readable patterns. The comparison between the two datasets (sukuma wiki vs. maize) activates analogical reasoning, helping learners generalise the tool rather than seeing it as one-off procedure.	Observable indicator: Learners correctly identify KES 10 as the modal price (frequency 6), state that prices exceeded KES 15 on 9 days, and offer a market-relevant interpretation (e.g., 'She should buy on days priced at KES 10, which happened most often'). Exit check: teacher scans the self-constructed maize frequency tables — a correctly built table will have rows for values 4, 8, 9, 12, 14, 17, 22, 25, 31 with frequencies 4, 1, 3, 2, 2, 3, 2, 1, 2 (total = 20).

		— record as a DQB question for Lesson 3 on grouped frequency tables.) Close by asking: 'Rewrite the raw maize list from the board into a frequency table in your exercise book. You have 5 minutes.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Regrouping whole number place values](#) Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos](#) 📖 **HTML:** [Probability Distribution (Flexbook)](#) Source: CK-12 [🔍 Search ARES for similar readings](#)	Learners return to the class Driving Question Board. Each learner receives two sticky notes (or writes in a designated DQB section of their exercise books if physical notes are unavailable). On the first note (green/evidence): they write one thing the frequency table NOW lets them see about the maize harvest data that they could NOT see from the raw list. On the second note (orange/question): they write one new question that has emerged — something the frequency table still does NOT tell them about the Nakuru farmers. Learners place/write their notes and the teacher facilitates a 5-minute whole-class read-aloud of selected entries, grouping similar questions visibly on the board.	Say: 'Our Driving Question is: how can we collect, organise, and represent data to help Kenyan decision-makers? Today we gave ourselves a new tool. Let's update our board.' Distribute sticky notes or direct learners to the DQB section. Say: 'Green note: what can you NOW see in the maize data thanks to a frequency table? Orange note: what does the table still NOT tell you — what question does it leave unanswered?' Allow 4 minutes of individual writing. Collect/post notes. Read aloud 4–5 green notes, then 4–5 orange notes. As you read orange notes, say: 'Is this question one we should track? Raise your hand.' For questions that get 5 or more hands, star them on the DQB as 'priority questions.' Likely emerging questions: 'Which harvest number is most typical?' / 'What if we added all the numbers and divided?' / 'What about farmers with really unusual harvests — do they throw off the picture?' Say: 'These questions will drive our next lessons. Hold onto them.'	Driving Question Board as Epistemic Tracker: The DQB makes the class's collective knowledge growth visible and public. Separating 'evidence' notes from 'question' notes trains learners to distinguish what they know from what they still need to find out — a core scientific practice. Starring priority questions democratises the inquiry direction and increases learner ownership of the storyline.	Observable indicator: Green notes demonstrate specific, table-grounded observations (e.g., 'I can now see that the value 4 bags appeared most often — 4 times' rather than vague statements like 'it is clearer'). Orange notes articulate genuine mathematical gaps (e.g., 'the table doesn't tell me what the middle harvest is'). Teacher flags any green notes that are too vague for targeted follow-up in Lesson 3.
Model Building Phase 📹 **VIDEO:** [Regrouping whole number place values](#) Source: Khan Academy (English - US curriculum)	Learners return to the predictive model they drew in Lesson 1 (a sketch or written explanation of how they thought the KALRO officer could find the 'one best number' for the Nakuru County harvest). They now revise or annotate their model to incorporate today's learning. Specifically,	Say: 'In Lesson 1, you drew a first model of how to find the best number to represent all 20 Nakuru farmers. Get that model out now.' Allow 1 minute to locate models. Say: 'Today we learned that Step 1 must always be: organise the raw data. Add that to your model now. Show what a frequency table	Cumulative Model Revision: Requiring learners to physically annotate and expand their Lesson 1 model (rather than starting a new one) makes conceptual growth tangible. The 'glow and grow' peer protocol introduces constructive critique as a mathematical practice, mirroring how real data	Observable indicator: Revised models clearly show a sequential process (Step 1: build frequency table → reveals: most common value, spread → does NOT reveal: mean, median) rather than jumping directly to calculating an average. Teacher collects 5–6 models at random to review before

🔍 Search ARES for similar videos 📖 **HTML:** Probability Distribution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	learners add a 'Step 1' to their model: 'Before finding any representative number, first organise the raw data into a frequency distribution table.' They label what the table reveals (most common value, how spread out the data is) and note what it still does not reveal (the mean, the median). Learners share revised models with a new partner (not their pair from the Observe phase) and give each other one 'glow' (something strong in the model) and one 'grow' (something to improve or add).	reveals AND what it leaves unanswered. You have 6 minutes.' Circulate — prompt with: 'Is the most common value always the best representative number? Add that question to your model.' After 6 minutes, say: 'Find a partner you haven't worked with today. Share your model. Give them one glow — one thing their model does really well — and one grow — one thing they should add or change. Be specific. You have 4 minutes total, 2 minutes each.' Cold-call 2 pairs after sharing: 'What was the most useful grow your partner gave you?' Allow 10 seconds WAIT TIME. Close the lesson by saying: 'Our model is growing. By Lesson 9, it will answer the officer's question completely. Today, we gave it its foundation: organised data.'	scientists refine analytical frameworks through peer review. Connecting the revised model explicitly to the KALRO officer's problem sustains the phenomenon as the anchor for all learning.	Lesson 3 — look for learners who still show no step-by-step structure as candidates for small-group support.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners' tally charts and frequency tables all sum to 20 — and for those that did not, what misconception drove the error (double-counting, skipping values, wrong grouping)? (2) Were the green DQB notes specific and table-grounded, or vague? If vague, how will you model stronger evidence claims at the start of Lesson 3? (3) Which learners appeared disengaged during the pair work — were they struggling with the arithmetic of tallying, or with the concept of frequency as a count? Plan differentiated entry points for Lesson 3 accordingly. (4) Did the sukuma wiki context feel authentic and engaging, or did learners need more grounding in the Kongowea Market setting? Consider beginning Lesson 3 with a 2-minute image or brief description of Kongowea Market to deepen context. (5) Did the DQB orange notes surface the idea of 'most common value' (mode) independently? If so, Lesson 3 can begin by honouring that student-generated question as the entry point into measures of central tendency.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners watched a KALRO data clerk build a frequency distribution table from a raw maize harvest list, and then constructed their own tally charts and frequency tables using daily sukuma wiki prices from Kongowea Market, Mombasa (20 market days, prices ranging from KES 10 to KES 25).
What did I learn?	A frequency distribution table organises raw data by listing each value alongside how many times it occurs (its frequency), using tally marks as a systematic counting tool. A well-structured table makes patterns — such as the most common value and the spread of data — immediately visible in a way that a raw list does not.
How does this explain the	Learners connected frequency tables to the Nakuru Maize Harvest Mystery by constructing a frequency table for the 20 raw

phenomenon?	harvest values, identifying what the table now reveals (e.g., the value 4 bags appeared most often) and what it still does not answer (e.g., which single number best represents all farmers — setting up the need for mean, median, and mode in upcoming lessons).

LESSON 3 (80 minutes): Finding the Balance Point: Mean of Ungrouped Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students will calculate the mean of ungrouped data and evaluate whether the mean is the best representative value for the Nakuru County maize harvest dataset.
Knowledge	The mean (arithmetic average) of ungrouped data is calculated by summing all values and dividing by the total number of values. The mean represents the 'balance point' of a dataset — the point at which the data would perfectly balance if placed on a number line. Outliers can pull the mean away from the centre of most values, making it potentially unrepresentative of a typical observation.
Skills	Students will: (1) calculate the mean of ungrouped data sets by hand and using the PhET 'Mean and Median' simulation; (2) locate the mean on a dot plot and number line; (3) compare the computed mean to the spread of the Nakuru harvest data; (4) construct a written argument about whether the mean is the best single representative number for the farmers' data.
Attitudes	Students demonstrate responsibility by recognizing that the choice of a representative number has real consequences for Kenyan farmers who depend on subsidized fertilizer allocations. They show diligence and integrity when computing and cross-checking their calculations.
Key Inquiry Question	How do we collect, organize, and represent data? How do measures of central tendency help us make sense of data in real life?
Purpose in Storyline	In Lessons 1 and 2, students organized the raw Nakuru harvest data into frequency distribution tables and began to notice where data clusters. In Lesson 3, students investigate the mean as a mathematical 'balance point' using the PhET simulation and apply it to both the maize harvest data and a mandazi sales dataset. They return to the anchoring phenomenon question — 'Which single number best represents all our farmers?' — and evaluate whether the mean earns that title, setting up the comparison with median and mode in Lessons 4 and 5.
Safety Notes	No physical hazards. Ensure responsible, focused use of devices running the PhET simulation. If using shared tablets, remind students to handle devices with care. Students with visual difficulties should be seated closer to any projected simulation display.

B. LESSON OVERVIEW

Lesson 3 opens by reconnecting to the Nakuru Maize Harvest Mystery. Students first make a written prediction about what a 'fair share' harvest would look like if all bags were redistributed equally among the 20 farmers. They then explore the PhET 'Mean and Median' simulation to build an intuitive, visual understanding of mean as a balance point before formalising the algorithm. Working in groups, they calculate the mean of the Nakuru harvest data (ungrouped) and a secondary mandazi sales dataset. The lesson closes with students evaluating whether the mean truly represents the 'typical' Nakuru farmer — and posting new questions on the Driving Question Board as they notice the mean can be distorted by unusually high or low values.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Ungrouped Data	Students receive a printed strip showing the 20 Nakuru farmers' maize harvests in the same	Distribute the data strips and read the redistribution scenario aloud. Say: 'Before we do any maths, I	Prediction with Justification — by requiring students to articulate a reason for their guess, this	Observable indicator: All students have a written number and a one-sentence justification on their strip

to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	unordered form from the anchoring phenomenon video (e.g., 4, 31, 12, 18, 7, 25, 14, 9, 22, 11, 16, 28, 6, 19, 13, 10, 20, 5, 17, 24 bags). They read the scenario: 'The county officer wants to redistribute all the bags equally among the 20 farmers so every farmer gets the same share — a perfect balance. Without calculating, write your best guess for that equal-share number and explain your reasoning in one sentence.' Students write individually for 3 minutes, then share with a shoulder partner for 2 minutes. The teacher cold-calls 4 students to share predictions and records the range of guesses on the board under the heading 'Our Predictions.'	want your gut instinct. If you could scoop up all the bags from every farmer and share them out equally, how many bags would each farmer get? Write your number and one reason — go.' Allow 3 minutes of silent individual writing. After 2 minutes of pair discussion, cold-call exactly 4 students using name cards: 'Tell me your prediction AND your reason — one sentence.' Record each prediction on the board. Do NOT confirm or deny any prediction. Say: 'Hold onto these guesses — we are going to find out who was closest and why.' Wait 12 seconds after posting the last prediction before transitioning, allowing students to compare predictions silently.	strategy activates prior intuitive knowledge about fairness and sharing (equal distribution), creating a cognitive anchor that the formal algorithm of the mean will later satisfy or challenge. The visible range of class predictions creates productive tension and a reason to investigate.	before pair-share begins. Listen for reasoning types — 'I picked the middle,' 'I added a few and guessed,' 'I picked the most common' — to gauge whether students are confusing mean, median, and mode conceptually at the outset. Note which students predict values close to the actual mean (15 bags) vs. those anchored to the minimum (4) or maximum (31). This informs targeted questioning during Phase 3.
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students open the PhET 'Mean and Median' simulation (projected for the class if devices are limited; pairs share one device if available). The teacher guides a structured exploration in two rounds. Round 1 (8 min): Students drag data points on the simulation's dot plot and observe how the mean marker (blue triangle) moves. They record observations: 'What happens to the mean when I add a very high value? A very low value? What happens when all values are equal?' They sketch two dot-plot configurations in their exercise books and mark where the mean lands. Round 2 (7 min): Students switch to the 'Balance Point' visual mode in the simulation (or the teacher demonstrates on the projector), where the mean is shown as the fulcrum of a balance beam. Students observe that the beam only levels when the fulcrum is placed at the mean. They	Before launching the simulation say: 'We are going to use a simulation to SEE what the mean does — not just calculate it. Your job is to be a scientist: observe, record, and look for patterns.' Project the PhET simulation. For Round 1, say: 'Drag one point all the way to the right — to 30. What happens to the blue triangle? Write it down.' Wait 10 seconds. Cold-call 2 students. Then say: 'Now drag it back to 5. What changes?' Wait 10 seconds. Cold-call 2 different students. For Round 2 transition, say: 'Now switch to balance-beam mode. I want you to tell me — in one sentence — what the balance point is showing us about the data.' Wait 15 seconds for writing. Cold-call 3 students. During the live Nakuru data entry, pause before revealing the mean and say: 'Based on what you have seen, where do you predict the triangle will land? Point to the number line on your strip.' Scan	Interactive Simulation with Targeted Observation Prompts — the PhET simulation externalises the abstract concept of mean as a physical balance point, making the mathematics visually and kinaesthetically concrete. By providing specific observation prompts ('what happens when...') rather than open exploration, the teacher focuses students' attention on the mathematical properties of the mean (sensitivity to extreme values, equal redistribution) that are central to the lesson's conceptual goal.	Observable indicator: Students' exercise-book sketches show at least two dot-plot configurations with the mean triangle correctly positioned relative to the data spread. Listen for the language students use — do they say 'average,' 'middle,' or 'balance point'? Students who conflate mean with median (placing the triangle at the physical centre of the number line rather than the data's balance point) need a follow-up prompt: 'Is the balance point always exactly in the middle of the number line? Show me a case where it isn't.'

	record: 'In your own words, what does the balance point tell us about the mean?' Students then watch a 2-minute teacher-narrated demonstration using the Nakuru data: the teacher enters the first 10 harvest values into the simulation live and asks students to predict — before pressing 'show mean' — where the blue triangle will land.	the room to observe pointing. Then reveal. Say: 'Were you surprised? Why or why not?' Wait 12 seconds before taking responses.		
Explain Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Graphic Displays of Data (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formalises the mean algorithm: Mean = (Sum of all values) ÷ (Number of values), writing it on the board in both symbolic form ($\bar{x} = \Sigma x \div n$) and plain language. Students work in groups of four to calculate the mean of the 20 Nakuru harvest values from their data strips, showing all working. One student in each group acts as 'calculator,' one as 'recorder,' one as 'checker' (re-adds the sum independently), and one as 'reporter.' Groups then receive a second dataset: daily mandazi sales (in units) recorded by a street vendor in Gikomba Market, Nairobi over 10 days: 34, 50, 42, 38, 55, 41, 200, 39, 44, 37. They calculate the mean and then discuss: 'Does this mean accurately represent a typical day of mandazi sales? Why or why not?' Each group writes a two-sentence conclusion connecting back to the anchoring question: 'Does the mean represent the typical Nakuru farmer?' Groups share conclusions; the teacher records key phrases on the board.	Write the formula and say: 'This is the mathematical way of doing exactly what the simulation showed — finding the balance point. Let's prove it with our own Nakuru data.' Assign roles explicitly: 'Person nearest the window: calculator. Next person: recorder. Next: checker. Person nearest the door: reporter.' Circulate and check that the 'checker' is independently re-summing, not just confirming. After groups finish the Nakuru calculation, say: 'Before I tell you the answer — reporters, stand up and tell me your group's mean. Everyone else, give a thumbs up if they match yours.' Wait 10 seconds for comparison. Address discrepancies by asking: 'Two groups have different answers — what might have gone wrong? How do we find out?' After confirming the Nakuru mean (15 bags), distribute the mandazi dataset and say: 'Calculate the mean — but this time, after you get the number, ask yourselves: does this number make sense as a typical day?' Wait 15 seconds after groups finish calculations before calling for responses. Cold-call 3 reporters to share their two-sentence conclusions. On the board, write any phrases mentioning 'the 200' as an outlier	Structured Group Roles with Contrasting Cases — assigning computational roles ensures accountability and distributes cognitive load, so all students engage with the arithmetic. The 'contrasting case' of the mandazi dataset (which contains the outlier 200) is deliberately chosen to create cognitive dissonance: the mean of the mandazi data (~58) is higher than 9 out of 10 actual days, making it visibly unrepresentative. This contrast sharpens students' understanding of when the mean is — and is not — trustworthy, directly advancing the driving question.	Observable indicator: Each group's written working shows the summing step and division step separately (not just a final answer). Reporters' two-sentence conclusions explicitly mention either (a) the outlier's effect on the mean, or (b) a comparison between the mean and the actual individual values. A student who cannot explain why the mandazi mean seems 'too high' needs the prompt: 'Circle the value that is most different from all the others. If you removed it, what would happen to the mean? Try it.'

		and label it: 'This is what mathematicians call an OUTLIER — a value that is far from the others. How might an outlier affect our Nakuru mean?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Graphic Displays of Data (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students individually write on a sticky note (or the teacher writes class contributions on the board in two colours): (1) GREEN — one thing they now know that helps answer the driving question ('How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?'); and (2) ORANGE — one new question they now have that the class has not yet answered. The teacher reads out selected sticky notes and places them on the class DQB under two columns: 'Evidence We Have Gathered' and 'Questions Still to Investigate.' The class briefly discusses whether the mean alone is enough for the county officer to make her fertilizer decision — or whether another measure might do better.	Distribute two sticky notes (or two colours of paper) and say: 'Green note: write one thing you now KNOW that helps us answer our driving question. Orange note: write one new question our lesson today has raised for you — something we have NOT yet answered.' Allow 3 minutes of silent writing. Collect and read 5–6 notes aloud, choosing a mix of strong conclusions and genuine new questions. Place them on the DQB visibly. Point to the orange questions and say: 'These questions — about the middle value, the most common value, what happens with outliers — are exactly what we will investigate in Lessons 4 and 5. Keep them in mind.' End with the anchoring question: 'Right now, if you were the KALRO officer, would you use the mean of 15 bags to recommend fertilizer for every farmer in Nakuru? Turn and tell your partner — yes, no, or it depends — and give one reason.' Wait 12 seconds for pair talk, then cold-call 3 students.	Driving Question Board Update with Evidence Sorting — the DQB is a visible, cumulative record of the class's growing understanding. Distinguishing 'evidence gathered' from 'questions still open' models scientific inquiry practice (NGSS SEP 1: Asking Questions; SEP 6: Constructing Explanations) and ensures students see each lesson as one piece of a larger investigation rather than an isolated topic. The turn-and-talk on the fertilizer decision reconnects mathematics to real Kenyan community impact.	Observable indicator: Green sticky notes reference the mean formula, the balance-point concept, or the outlier effect — not vague statements like 'I learned about averages.' Orange sticky notes pose genuine mathematical questions (e.g., 'What if we used the middle number instead?' or 'What do we do when there is no one most-common value?'), indicating readiness to engage with median and mode in subsequent lessons. Students who cannot generate an orange question are asked: 'You found that 200 mandazi pulled the mean up — what other number could you use instead to represent a typical day?'
Model Building Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12 🔍 Search ARES for similar videos	Students return to their 'Data Story Model' — a cumulative visual model started in Lesson 1 that represents their current best understanding of how to make sense of the Nakuru harvest data. In Lesson 1 they added the raw data list; in Lesson 2 they added a frequency distribution table. Today, each student adds: (a) the calculated mean (15 bags) marked	Display a teacher model on the projector showing Lessons 1 and 2 layers already added. Say: 'Your model is a living document — every lesson we add a new layer of understanding. Today's layer is the mean.' Walk through the three additions on the projected model: 'First, mark 15 on your number line and draw a blue triangle. Second, write the equal-redistribution	Cumulative Model Revision (Layered Data Story Model) — requiring students to physically add each new concept to a single evolving model reinforces that statistics is a coherent toolkit, not a collection of isolated procedures. The red-asterisk warning note makes metacognitive awareness (knowing when a tool is limited) a visible, permanent part of the	Observable indicator: The blue triangle is correctly positioned at 15 on the number line (not at the centre of the axis). The equal-redistribution sentence is complete and mathematically accurate. The red asterisk note references either 'outlier,' 'extreme value,' or 'unequal spread' — not just a vague 'it might be wrong.' Students who position the triangle at the

📖 **HTML:** [Graphic Displays of Data (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	clearly on their number line with a labelled blue triangle (to match the PhET balance-point symbol), (b) a one-sentence annotation: 'The mean tells us that IF all 20 farmers shared their harvest equally, each would receive ___ bags,' and (c) a red asterisk next to the mean with a note: 'Warning — one very high or very low value can shift the balance point. Is 15 bags really typical?' Students then update their model's title banner to read: 'What I know so far about the Nakuru harvest data' and add today's date. In the final 3 minutes, pairs compare models and check each other's mean annotation for accuracy.	sentence — fill in the blank. Third — and this is the critical thinking step — add the red asterisk warning. Mathematics is not just calculation; it is also knowing when to trust your calculation.' Circulate during the 5-minute model update. Prompt students who draw the triangle in the wrong position: 'Is 15 closer to the minimum of 4 or the maximum of 31? Does your triangle show that?' For the pair-check, say: 'Swap models. Does your partner's blue triangle sit at 15? Does their warning note mention outliers or unequal distribution? If not, help them revise.' Close by saying: 'Next lesson, we will add the median — the middle value — to this same model. You will then be able to see both the balance point AND the midpoint on the same number line. Will they always be in the same place? That is your overnight question.'	model, directly countering the misconception that the mean is always the correct choice. This builds toward the lesson sequence's culminating task of advising the county government.	numerical midpoint of 4 and 31 (i.e., at 17.5) reveal a persistent mean/median confusion and should be flagged for targeted support at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the PhET simulation, did students articulate the balance-point concept in their own words, or did they default to describing the mean procedurally as 'add and divide'? If students struggled with the visual metaphor, consider beginning Lesson 4 with a physical balance-beam demonstration using stacks of exercise books to reinforce the concept before introducing the median. (2) When groups encountered the mandazi dataset with the outlier (200 units), did they spontaneously question the mean's representativeness, or did they accept it uncritically? If most groups did not flag the outlier, the contrast case strategy did not fully achieve its purpose — consider making the outlier discussion more explicit at the start of Phase 3 in Lesson 4. (3) Review the orange DQB sticky notes: do students' new questions naturally lead toward median and mode, or are there unexpected conceptual gaps (e.g., confusion about what 'sum' means, difficulty distinguishing number of farmers from number of bags)? Use these questions to adjust the opening of Lesson 4. (4) Check three to five students' Data Story Models for the correct placement of the blue triangle — this is the most reliable formative indicator of whether the balance-point concept has landed. Students who placed the triangle at 17.5 (midpoint of range rather than mean of data) need a one-on-one clarifying conversation before Lesson 4 introduces the median, or the two concepts will compound into deeper confusion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students explored the PhET 'Mean and Median' simulation and observed how the mean marker (blue triangle) shifts as data points are added, removed, or dragged to extreme values. They observed the balance-beam visual mode and recorded what happens to

	the mean when an outlier (e.g., 200 mandazi sales) is introduced into a dataset.
What did I learn?	The mean is the arithmetic average calculated by summing all values and dividing by the number of values ($\bar{x} = \Sigma x \div n$). It represents the 'balance point' of a dataset — the equal-redistribution value. For the 20 Nakuru maize harvest values, the mean is 15 bags. However, outliers (extreme values far from the rest of the data) can pull the mean away from what is truly typical, as seen in the Gikomba mandazi sales data where one day of 200 sales distorted the mean.
How does this explain the phenomenon?	Students explained in two-sentence group conclusions whether the mean of 15 bags accurately represents the typical Nakuru farmer, connecting the balance-point concept to the anchoring phenomenon. They updated their cumulative Data Story Models with the mean marked on the number line, an equal-redistribution annotation, and a critical 'warning' note about outliers — building the conceptual foundation for comparing mean, median, and mode in Lessons 4 and 5.

LESSON 4 (40 minutes): The Most Popular Value: Mode of Ungrouped Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to identify and calculate the mode of ungrouped data and evaluate whether mode or mean is a more useful representative value for real-world decision-making.
Knowledge	Learners will know that the mode is the value that appears most frequently in a data set, that a data set can have one mode (unimodal), two modes (bimodal), or no mode, and that the mode is not affected by extreme outlier values.
Skills	Learners will be able to collect classroom data using tally marks, construct a frequency distribution table, identify the mode from the table, and compare the mode with the previously calculated mean to argue which better represents a given data set.
Attitudes	Learners will appreciate that different statistical measures serve different purposes and that choosing the right measure is a critical thinking skill that affects real decisions for communities like Nakuru County farmers.
Key Inquiry Question	How do we collect, organize, and represent data? How do measures of central tendency help us make sense of data in real life?
Purpose in Storyline	In Lesson 3, learners discovered that the mean can be distorted by outliers. Lesson 4 introduces mode as an alternative measure that reflects the most common experience — the value most farmers actually reported. This advances the driving question by giving learners a second tool to recommend a representative harvest value to the Nakuru County government, while also surfacing the important idea that no single measure is always best.
Safety Notes	No physical safety concerns. Ensure inclusive participation during the classroom data-collection activity — sensitivity is required if personal questions could embarrass learners. The number-of-siblings question is generally safe but the teacher should offer an opt-out, allowing any learner to write their value privately rather than call it out.

B. LESSON OVERVIEW

Learners begin by predicting which harvest value appears most often in the Nakuru maize data, connecting to the anchoring phenomenon. They then collect live classroom data on number of siblings, organize it into a frequency table, and identify the mode. An analysis of the mandazi market supporting phenomenon is used to consolidate the concept. Learners debate whether mode or mean better serves the Nakuru County fertilizer decision, and they update the Driving Question Board with new evidence. The lesson is structured around the Predict-Observe-Explain framework to keep learners doing the thinking.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Ungrouped Data to Find the Mean Principles] Source: CK-12	Learners are shown the 20 Nakuru maize harvest values on the board (4, 8, 12, 15, 15, 15, 10, 7, 20, 18, 15, 9, 22, 15, 11, 31, 6, 15, 14, 10) — the same data set from Lessons 1 and 3. The teacher asks:	Display the 20 values clearly on the board or a printed handout. Say: 'You already know the mean of this data is 15 bags. Today we are going to explore a completely different way of finding a	Prediction journaling activates prior knowledge and creates cognitive investment. Recording predictions publicly on the board creates accountability — learners will want to check whether they	Circulate during the 90-second individual writing period. Check that learners are engaging with the actual data values rather than guessing randomly. Listen for reasoning that references

🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Without doing any calculation, which single harvest number do you think appears most often in this list? Write your prediction in your notebook and give one reason for your choice.' Learners write individually for 90 seconds, then share predictions with a shoulder partner. Two or three pairs share with the class. The teacher records predicted values on the board without confirming or denying correctness.	representative value — one that does not involve any addition or division at all. Before I tell you anything, I want your brain to make a guess.' After individual writing time, say: 'Turn to your partner and compare predictions. Do you agree? Why?' After pair discussion, cold-call two pairs: 'Amina, what did you and your partner predict, and what was your reasoning?' WAIT TIME: After asking each pair, wait at least 5 seconds before accepting the answer. Record all predicted values on the board in a column labeled 'Our Predictions.' Do not evaluate yet — say: 'Let us keep these predictions and see what the data actually tells us.'	were right during the Observe phase. Connecting to the mean (already known) primes learners to think comparatively from the start.	frequency (e.g., 'I see 15 written several times') — this signals readiness for the concept. Note learners who look confused so you can pair them strategically during the activity.
Observe Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Classroom Data Collection (8 minutes): The teacher explains that the class will collect its own data set right now. Each learner counts the number of siblings they have (including half-siblings and step-siblings if comfortable) and writes the number on a small piece of paper or calls it out when asked. The teacher records each value on the board as it is called, in the order given, creating a raw unsorted list of approximately 35 values. Learners copy this raw list. Part B — Building the Frequency Table (5 minutes): In pairs, learners organize the raw list into a frequency distribution table with three columns: Number of Siblings, Tally, and Frequency. They identify which value has the highest frequency. Part C — Mandazi Market Check (4 minutes): The teacher reveals the mandazi daily sales data from the supporting phenomenon (values:	For Part A, say: 'I need everyone to participate. If you prefer not to call out your number, just hold up fingers or write it on paper and I will read it. No one is excluded.' Record values clearly and number them so learners can track the count. After collecting all values say: 'Look at this list. Does it remind you of anything?' (Expected: the chaotic raw Nakuru data from Lesson 1.) For Part B, say: 'With your partner, do exactly what we did in Lesson 1 — build a frequency table. Then circle the row with the highest frequency. That is your answer.' WAIT TIME: Give pairs 4 full minutes without interruption. Resist the urge to help immediately. After 4 minutes, ask: 'What value appears most often in our class data?' Take 3 responses. For Part C, reveal the mandazi data and say: 'You have 90 seconds — build the table, find the most frequent value, done.' After 90 seconds, say: 'What did	Using live classroom data makes the concept personally relevant and memorable — learners are not just analyzing numbers, they are analyzing themselves. The three-part sequence (raw list, frequency table, mode identification) mirrors exactly the process used for the Nakuru data, reinforcing the lesson storyline. Juxtaposing the mandazi data with the Nakuru data helps learners generalize the concept beyond a single context.	Observe frequency tables being constructed during Part B. Look for: correct tally marks, accurate frequency counts, and correct identification of the modal value. Common error to watch for: learners who identify the highest value in the data (e.g., 31 bags) rather than the most frequently occurring value. If this error appears, ask without correcting: 'What does most frequent mean — biggest, or appearing the most times?'

	45, 50, 45, 60, 45, 55, 45, 70, 45, 50, 45, 80, 45, 55, 45). Learners quickly build a small frequency table and identify the mode. The teacher then asks: 'Is the mode the same as the mean we calculated in Lesson 3 for a similar data set?'	you find?' WAIT TIME: 4 seconds after the question before taking responses.		
Explain Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formalizes the concept of mode. Learners record the definition, notation, and rules in their notebooks. The class then works through three examples together: (1) finding the mode of the class siblings data, (2) finding the mode of the Nakuru maize harvest data (confirming or refuting the Predict phase predictions), and (3) a bimodal example using a short data set of mandazi prices from two Nairobi market stalls: 10, 5, 10, 8, 5, 10, 5, 12. Learners discover that both 5 and 10 each appear three times — introducing the idea of a bimodal data set. Finally, learners are given a 5-value data set (2, 4, 6, 8, 10) and asked whether a mode exists, introducing the concept of no mode. After the three examples, the teacher leads a structured debate: 'For the Nakuru County fertilizer decision — which is more useful, the mean (15 bags) or the mode (15 bags in this case)? Would your answer change if the mode were 10 bags instead of 15?'	Write on the board: 'The MODE is the value that occurs MOST FREQUENTLY in a data set.' Say: 'Mode comes from the French word meaning fashionable — it is the most popular value, the one that shows up to the party the most.' For the Nakuru data, say: 'Let us check our predictions from the beginning of class.' Count frequencies together. When 15 is confirmed as the mode, ask: 'Interesting — the mean is also 15. Is that always the case? Will mode and mean always be equal?' WAIT TIME: 6 seconds. Accept responses without confirming. For the bimodal example, say: 'What happens when two values tie for most frequent? Talk to your partner for 30 seconds.' After sharing, introduce the term bimodal. For the no-mode example, ask: 'Can a data set have NO mode? When would that happen?' WAIT TIME: 5 seconds. For the debate, use a structured protocol: give learners 60 seconds to write their argument privately, then have three learners argue for mean and three argue for mode. Conclude by saying: 'Both are valid tools. The skill is knowing when each one serves the decision-maker better.'	The French etymology of mode creates a memorable hook. The prediction-checking ritual (returning to the board predictions) closes the cognitive loop opened at the start, rewarding learners who reasoned from frequency. The structured debate is critical — it advances the driving question by forcing learners to articulate a position on which measure best serves the Nakuru County government, using evidence from their own calculations.	Use a quick exit check: ask learners to hold up 1 finger if the data set has one mode, 2 fingers if bimodal, and a fist if no mode, as you display three mini data sets on the board. This gives instant whole-class feedback before the DQB phase. Listen during the debate for learners who can articulate the idea that mode is resistant to outliers — this is the key conceptual leap linking Lessons 3 and 4.
Driving Question Board (DQB) Creation	Learners return to the class Driving Question Board. The teacher reveals the current state of the board, including questions and	Say: 'In Lesson 3, someone asked on the DQB whether there is a measure that is not affected by outliers like the 200-mandazi day.	The DQB functions as a public knowledge artifact — it externalizes thinking and shows learners that understanding is	Read sticky notes quickly as they are posted. Green notes should reference mode, frequency, or a comparison with mean. Yellow

📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	claims posted in Lessons 1 through 3. Learners are given two sticky notes each. On the first (green), they write one new piece of EVIDENCE they have gathered today that helps answer the driving question: 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' On the second (yellow), they write one new QUESTION or TENSION that today's lesson created — for example, 'What happens when the mode and mean give different answers?' or 'Is mode useful for the fertilizer decision if most farmers harvested a very low number?' Learners post their sticky notes in the appropriate columns on the DQB. The teacher reads three or four aloud and briefly connects them to upcoming lessons.	Today you found one — the mode. Let us add that evidence.' Point to relevant previous sticky notes. Give learners 3 minutes of quiet writing time for their two sticky notes. As they post, circulate and read notes. Select two or three that represent contrasting views and read them aloud: 'Listen to what Wanjiku wrote — she says mode is better because it shows what most farmers actually experienced. But Kipchoge wrote that if the most common harvest is very low, using it as the representative value might cause the government to under-allocate fertilizer. Both are valid tensions.' Say: 'Lessons 5, 6, 7, and 8 will give us more tools to resolve these tensions. Keep your questions alive.'	cumulative. By explicitly linking today's mode findings to a question posted in Lesson 3, the teacher models how scientists and decision-makers build on prior evidence. The green-and-yellow sticky note protocol distinguishes between evidence (what we know) and questions (what we still need to find out), mirroring scientific inquiry practice.	notes should contain a genuine tension or unresolved question — if a learner writes a factual question with an obvious answer, prompt them: 'You already know that answer — what is still puzzling you?' This pushes deeper thinking. Count the ratio of evidence notes to question notes: if questions far outnumber evidence notes, the class may not have consolidated the concept and the teacher should open Lesson 5 with a brief mode review.
Model Building Phase 📹 **VIDEO:** Ungrouped Data to Find the Mean Principles Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work individually to complete a structured model-building task in their notebooks titled: 'My Statistical Toolkit — Updated After Lesson 4.' They draw a two-column comparison table with the headers MEAN and MODE. For each measure they fill in: (a) how to calculate it, (b) one advantage, (c) one disadvantage, and (d) one situation from Kenya where this measure would be the best choice. They then write a single recommendation sentence addressed to the fictional Nakuru County Agricultural Officer: 'For the fertilizer allocation decision, I recommend using [MEAN / MODE] because...' Finally, learners calculate the mode of one homework data set: the number of customers served daily at a	Say: 'You now have two tools in your statistical toolkit — mean from Lesson 3 and mode from today. A good statistician does not just know how to calculate both — they know when to use each one. That is what this model-building task is testing.' Circulate during the 5-minute task. For the comparison table, prompt learners who are stuck on disadvantages by asking: 'Think about the 200-mandazi day from Lesson 3 — which measure was affected by that outlier, and which was not?' For the recommendation sentence, look for learners who write a complete justification rather than a bare assertion. Select two contrasting recommendations to share with the class in the final 2 minutes: one recommending mean and one	The two-column model is a knowledge-organization tool that forces comparison rather than isolated recall. The recommendation sentence is a low-stakes performance task that mirrors the final Lesson 9 advisory report, building the skill incrementally. The homework problem uses a data set where the mode (8) is significantly lower than the mean (11.3), deliberately creating a tension that will motivate learners to want the median as a tiebreaker — priming curiosity for Lesson 5.	Collect or photograph three to five completed comparison tables at the end of the lesson. Check for: correct disadvantage for mean (affected by outliers), correct disadvantage for mode (ignores all other values, can be low or high), and a recommendation sentence that references the Nakuru context specifically rather than being generic. Use these as evidence for differentiation planning in Lesson 5 — learners who struggle with the disadvantages column need additional scaffolding on comparative thinking before the median is introduced.

	Kisumu fish market over 10 days (8, 12, 8, 15, 8, 20, 12, 8, 14, 8) and write whether they would use mode or mean to report a typical day to the market manager, giving one reason.	recommending mode, both with valid reasoning. Say: 'Both of these can be correct — it depends on the specific data and the specific decision. That is the complexity of real statistics.' Assign the Kisumu fish market problem as homework and say: 'In Lesson 5, you will get a third tool — the median — that will help you make an even stronger recommendation to the county government.'		

D. TEACHER REFLECTION
After the lesson, ask yourself: (1) Did learners clearly distinguish between the value that is largest and the value that is most frequent? This is the most common misconception — if it persisted, open Lesson 5 with a 2-minute targeted correction using a simple data set. (2) Did the structured debate produce genuine argumentation, or did learners simply agree with each other? If the debate lacked tension, consider assigning roles (devil's advocate) in future discussions. (3) Were learners able to connect mode to the anchoring phenomenon, or did the class activity data (siblings) become disconnected from the Nakuru story? If disconnection occurred, strengthen the bridging language between Part A and Part C of the Observe phase in future delivery. (4) Look at DQB sticky notes: do the questions posted reflect statistical thinking (about what the data means) or procedural confusion (about how to calculate)? Procedural confusion signals the need for more practice before Lesson 5; genuine statistical questions signal readiness for the median lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In the classroom siblings data and the mandazi market data, one value kept showing up more than all the others. In the Nakuru maize harvest data, the value 15 appeared 6 times out of 20 — more than any other value. In a bimodal data set, two values tied for most frequent. In a data set where every value is different, there is no mode at all.
What did I learn?	The mode is the value that occurs most frequently in a data set. It can be found by building a frequency table and identifying the row with the highest frequency. A data set can be unimodal (one mode), bimodal (two modes), or have no mode. Unlike the mean, the mode is not pulled away from the typical value by outliers. For the Nakuru maize data, both the mean and mode happen to equal 15 bags — but this is not always the case.
How does this explain the phenomenon?	The mode represents the most common experience in a data set, which is why it can be more useful than the mean when one typical value keeps repeating. For the Nakuru County fertilizer decision, the mode tells us what harvest level most farmers actually achieved — not an average that could be distorted by one farmer with an unusually large or small harvest. However, if the most common value is very low, using the mode might cause decision-makers to under-estimate the resources needed. This tension — and the need for a third measure — will be resolved in Lesson 5.

LESSON 5 (80 minutes): The Middle of the Story — Median of Ungrouped Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' ability to calculate and interpret the median of ungrouped data, and to compare mean, mode, and median as tools for representing a dataset — using real Kenyan contexts to determine which measure best serves decision-making.
Knowledge	Learners will know that the median is the middle value of an ordered dataset, that it is found differently for odd and even numbers of values, and that it is less affected by extreme values (outliers) than the mean.
Skills	Learners will be able to arrange ungrouped data in ascending or descending order, locate or calculate the median for both odd and even datasets, and compare mean, mode, and median to select the most representative measure for a given context.
Attitudes	Learners will appreciate the importance of choosing the right statistical tool when making decisions that affect real communities, demonstrating responsibility and a commitment to data-driven fairness.
Key Inquiry Question	How do measures of central tendency help us make sense of data in real life?
Purpose in Storyline	In Lesson 4, learners calculated the mean and mode for the Nakuru maize harvest data and noticed that a few very high or very low values pulled the mean away from what most farmers actually experienced. In Lesson 5, learners discover the median — the 'middle of the story' — and test whether it gives a fairer picture of the typical Nakuru farmer's harvest. Using KCPE scores from Kisumu as a supporting phenomenon, they see exactly when and why median outperforms mean, and they update their Driving Question Board with a more nuanced answer: the best single number depends on the shape of the data.
Safety Notes	No physical or chemical hazards. Ensure psychological safety: when discussing community data (harvests, exam scores), remind learners that real people are behind every number. Avoid language that shames low values.

B. LESSON OVERVIEW

Learners open by revisiting the Nakuru maize harvest data from earlier lessons and predicting whether the median or the mean better represents a 'typical' farmer. They then engage in a kinaesthetic ordering activity — physically lining up as 'data points' — to concretely discover how the median is located. A structured data investigation using KCPE scores from Kisumu schools (the supporting phenomenon) reveals the distorting effect of outliers on the mean and the median's resilience. Learners calculate median for both odd and even ungrouped datasets, then return to the Nakuru data to compare all three measures of central tendency (mean, mode, median) side by side. The lesson closes with a Driving Question Board update and a model revision in which learners annotate their statistical toolkit diagram with guidance on when to use each measure.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Video: Plinko Probability]	Learners are shown the Nakuru maize harvest data list (20 values, unordered) on the board — the same chaotic list the KALRO	Display the 20-value Nakuru dataset (unordered): 12, 4, 9, 21, 9, 17, 6, 31, 9, 14, 11, 8, 19, 9, 23, 7, 15, 28, 10, 9. Point to the mean	Prediction with Justification — By requiring a written sentence of reasoning before any instruction, learners activate prior knowledge	Circulate during individual writing and read 5–6 books. Look for: (1) learners who correctly anticipate that the median will sit between 9

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	officer wrote in the anchoring phenomenon video. They are also shown the mean (calculated in Lesson 4: 14.6 bags) and the mode (9 bags). The teacher poses a challenge: 'The county officer now hears about a third option — using the MIDDLE value of the ordered list. Before we calculate anything, predict: will the middle value be closer to the mean of 14.6 or the mode of 9? And do you think it will be a better representative of what a typical Nakuru farmer harvests?' Learners write individual predictions in their exercise books, justifying with at least one sentence of reasoning. Pair-share follows (90 seconds), then four learners share with the class. Predictions are displayed on a section of the chalkboard labelled 'OUR PREDICTIONS — We will test these.'	(14.6) and mode (9) already on the board from Lesson 4. Say: 'We have met two measures of central tendency — the mean and the mode. Today we investigate a third. But first — what does YOUR gut tell you?' Issue the written prediction prompt. After 3 minutes of silent individual writing, say: 'Turn to your neighbour — 90 seconds, share your prediction and your reason.' [WAIT 90 seconds.] Cold-call FOUR learners — deliberately pick two who look confident and two who look uncertain. Record all four predictions verbatim on the board. Say: 'We are scientists of data today. We do not delete wrong predictions — we test them. Let's find out.'	(ordering numbers, intuition about 'middle') and create a personal stake in testing their idea. This surfaces misconceptions early — for example, learners who confuse median with mean — so the teacher can address them explicitly during the Explain phase.	and 14.6 — this signals strong number sense; (2) learners who think median equals mean — flag for targeted questioning in Phase 3; (3) learners who give no reasoning — prompt with 'What do you already know about the word middle in mathematics?'
Observe Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — KINAESTHETIC ORDERING (15 min): The teacher selects 11 learners and gives each a card with one value from a simplified 11-value subset of the Nakuru data (4, 6, 7, 8, 9, 9, 12, 15, 17, 21, 28). Learners physically stand in a line across the front of the classroom in ascending order. The class directs them ('Move left! Move right!') until the line is correct. The teacher then asks: 'Without calculating anything — point to the person who is standing exactly in the middle.' The class identifies the 6th learner (value: 9). Teacher records: 'Median = 9 (for this 11-value set).' ACTIVITY B — SUPPORTING PHENOMENON — KISUMU KCPE DATA (15 min): The teacher projects or writes a scenario: 'A school inspector in	ACTIVITY A: Before selecting the 11 volunteers, say: 'I need 11 brave data points — yes, you will BE the data today.' Hand out pre-prepared number cards. Once lined up, say: 'Class — are they in order? Check your neighbour's work.' [WAIT 15 seconds for peer checking.] After the middle person is identified, ask: 'How many people are to their LEFT?' [Cold-call 1 learner.] 'How many to their RIGHT?' [Cold-call 1 different learner.] 'What do you notice?' [Cold-call 1 learner.] Record: 'For 11 values, median = the 6th value. Formula: position = (n+1)/2 = (11+1)/2 = 6th.' ACTIVITY B: Distribute the Kisumu KCPE scenario on a printed slip or write it clearly. Say: 'Work in pairs. You have 10 minutes. I want four things in your book: the ordered list, the	Kinaesthetic Modelling + Contrastive Cases — The physical line-up makes the abstract concept of 'middle value' concrete and spatial before any formula is introduced. The Kisumu contrastive case (same data with and without an outlier) is a high-leverage sensemaking strategy: by seeing the mean shift dramatically while the median barely moves, learners construct the concept of robustness directly from evidence rather than being told it.	During Activity A: check that learners directing the line are using 'ascending order' correctly — if they arrange in descending order, pause and ask the class: 'Does the order of direction matter for finding the median? Why or why not?' During Activity B: scan held-up books for the comparison table. Success indicators: (1) correctly ordered datasets; (2) median calculation accurate for odd n; (3) comparison table shows mean shifts by ~27 points with outlier, median shifts by 0–1 point. Flag pairs whose mean calculations are incorrect for re-teaching in Phase 3.

	Kisumu records the total KCPE marks for 7 learners from a pilot programme: 220, 234, 218, 229, 225, 231, 198.' Then a second scenario is added: 'One learner's score was entered incorrectly — it should be 412, not 198.' (This creates an outlier.) Learners, in pairs, (i) order both datasets, (ii) find the median of both, (iii) find the mean of both, and (iv) record in a two-column comparison table: 'With outlier / Without outlier — Mean vs Median.' They present findings on mini-whiteboards or exercise books held up for the teacher to scan.	median, the mean, and your comparison table. Go.' [Circulate, kneeling to pair level.] At 8 minutes, say: 'Two more minutes — start writing your conclusion sentence: The median changed by ___, the mean changed by ___.' After pairs finish, cold-call THREE pairs to share their comparison tables. Write the class consensus on the board: 'Mean with outlier vs without outlier' and 'Median with outlier vs without outlier.'		
Explain Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a structured notes framework (or copy from the board) that formalises what they discovered in Phase 2. The framework covers: (1) Definition of median; (2) Steps to find median — order data, find n, use position formula (n+1)/2 for odd n, average the two middle values for even n; (3) A worked example using the full 20-value Nakuru dataset (even n=20); (4) A side-by-side comparison table of mean, mode, and median for the Nakuru data with a column: 'What does this number tell the KALRO officer?' Learners then work individually on two practice problems: Problem 1 — find the median of a 9-value odd dataset of milk yields (litres/day) from dairy farmers in Kericho; Problem 2 — find the median of an 8-value even dataset of weekly market earnings (Ksh) for vendors at Gikomba Market, Nairobi. After individual work, learners compare answers with a partner and resolve any differences before a whole-class debrief.	Open the Explain phase by returning to the predictions on the board. Say: 'Let's check our predictions against what we just observed.' [Read each prediction aloud and mark ✓ or ✗ with the class — 2 minutes.] Then say: 'Now let's build the rule together.' Use guided questioning — do NOT lecture the rule unprompted. Ask: 'From our human data line, how did we find the position of the middle person?' [Cold-call 2 learners.] 'What formula did we write?' [Cold-call 1 learner.] 'What happens if we have an EVEN number of values — can there be ONE person in the exact middle?' [WAIT 15 seconds — cold-call 1 learner.] Guide learners to discover: average the two middle values. Work through the 20-value Nakuru dataset on the board step by step, narrating: 'Step 1 — I order all 20 values. Step 2 — n=20, positions are 20/2=10th and 11th. Step 3 — I find those two values and average them.' For practice problems, say: 'These are YOUR problems now. Silent	Guided Inquiry with Structured Notes + Prediction Revisit — Returning to predictions before formalising the rule creates a 'reveal moment' that consolidates learning emotionally and cognitively. The structured notes framework ensures all learners capture accurate content while the guided questioning keeps ownership of discovery with the learners rather than the teacher. The Kenyan practice contexts (Kericho dairy, Gikomba market) maintain the thread back to the driving question: data about real Kenyan livelihoods.	During individual practice: look for three specific errors — (1) failing to order data before finding the middle (most common error); (2) using n/2 instead of (n+1)/2 for odd datasets; (3) for even datasets, taking only the lower of the two middle values instead of averaging. During partner comparison: listen for productive disagreement — pairs who catch each other's errors are demonstrating metacognitive skill. During whole-class debrief: ask a learner who initially had a wrong prediction to explain what changed their thinking — this models intellectual flexibility.

		individual work — 8 minutes. No talking yet.' [Circulate, offering only Socratic prompts: 'What is n?', 'What position formula will you use?', 'Have you ordered the data first?'] After 8 minutes: 'Now — 3 minutes with your partner. If you agree, write a star. If you disagree, figure out who made an error.' Debrief with cold-calls: 3 learners for Problem 1, 3 learners for Problem 2.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Plinko Probability Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board and the two Key Inquiry Questions. In groups of four, they discuss and write a sticky note (or card) responding to the prompt: 'What can the KALRO officer now say about the Nakuru maize harvest data that she could NOT say after Lesson 4?' Each group also places a ✓ or ? next to each DQB question to indicate: 'We now have enough evidence to answer this' or 'We still need more evidence.' Groups share their sticky notes; the teacher facilitates a brief whole-class consensus on which parts of the driving question have been answered and which remain open. A new question that commonly emerges — 'When is median better than mean?' — is added to the DQB as a new investigable question for future lessons.	Transition by saying: 'We have now calculated mean, mode, AND median for the Nakuru data. Let's go back to the question that started everything.' Point to the Driving Question on the board. Give groups 5 minutes to write their sticky note — circulate and prompt with: 'Be specific — what number did you get for the median? How does it compare to the mean of 14.6? What story does that tell the officer?' After groups post their notes, facilitate a 5-minute gallery read: 'Read the notes around you silently — find one you agree with and one that surprises you.' [WAIT 2 minutes.] Cold-call 3 groups: 'What surprised you on someone else's note?' Synthesise at the board: draw a simple T-chart — 'Questions we can now answer / Questions still open.' Add any new learner-generated questions to the DQB. Say: 'These open questions are the engine of our next lessons.'	Driving Question Board Revision — The DQB is a public, cumulative record of the class's growing understanding. By explicitly marking questions as answered or still open, learners develop epistemic awareness — they can see the shape of their own knowledge and its gaps. The act of writing what the officer can NOW say forces learners to translate mathematical results into real-world meaning, reinforcing the purpose of statistics.	Read all sticky notes before the next lesson. Success indicator: groups correctly state that the median (approximately 10.5 bags for the Nakuru set) is lower than the mean (14.6) and that this suggests a few high-harvest farmers are pulling the mean up — making median a better representative of the majority. Flag groups whose notes focus only on calculation steps without interpretation — these groups need targeted questioning in the next lesson's opening.
Model Building Phase 📹 **VIDEO:** Video: Plinko Probability Source: PhET	Learners retrieve their 'Statistical Toolkit' diagram — a cumulative model they have been building since Lesson 3. In Lesson 3 they added the frequency distribution table; in Lesson 4, mean and mode. Today they add the median	Distribute or ask learners to open their Statistical Toolkit diagram. Say: 'This is your living model — it grows every lesson. Today it gets two new things: the median, and a decision guide for choosing which measure to use.' Give 10 minutes	Cumulative Model Revision with Decision Guide — The Statistical Toolkit is a learner-owned artefact that externalises growing conceptual understanding. Adding a decision flowchart elevates the model from a vocabulary list to a	Collect 5–6 models at random at the end of the lesson (photograph with a phone if physical books). Check for: (1) Correct median definition and formula; (2) Decision guide with at least two of the three conditions correctly mapped; (3) A

Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Graphic Displays of Data (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and, critically, a decision guide: a simple flowchart or annotated diagram answering 'Which measure should I use?' with three branches — (1) Is there an outlier? → Use median; (2) Is the data categorical or about most popular? → Use mode; (3) Is the data evenly spread with no outliers? → Use mean. Learners also annotate the Nakuru dataset summary on their model with all three measures labelled and a sentence explaining which measure they would recommend to the KALRO officer and why. Models are shared in pairs for peer feedback using two sentence starters: 'I can see that you understand… because…' and 'One thing to add or clarify is…'	for individual model revision. Circulate and prompt with: 'Where does median sit in your toolkit — is it separate from mean and mode, or connected?', 'Your decision guide — can a classmate use it without asking you any questions?', 'What would you tell the KALRO officer to do?' After 10 minutes, say: 'Pair up with someone who is NOT your usual partner. You have 4 minutes — use the two sentence starters on the board to give feedback.' [Write starters on board.] After peer feedback, give 2 minutes for learners to revise their model based on feedback. Close by cold-calling 2 learners: 'What is one thing you added to your model today, and why?'	reasoning tool — learners must think about the CONDITIONS under which each measure is appropriate, not just how to calculate it. Peer feedback using structured sentence starters ensures feedback is specific and academically focused rather than superficial.	clear recommendation for the KALRO officer with a reason referencing the Nakuru data. Learners who have only calculation steps and no decision guide will receive a written prompt in their book: 'Great calculation — now tell me: WHY median for this data?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the kinaesthetic line-up activity make the concept of median position genuinely concrete, or did some learners treat it as a game without connecting it to the formula? What adjustment — perhaps a second line-up with an even number of learners — would sharpen that connection? (2) When learners compared mean and median for the Kisumu KCPE outlier data, could they articulate in words WHY the median is more stable — or did they only observe the numerical difference? If interpretation was shallow, plan a structured 'convince the officer' writing task at the start of Lesson 6. (3) Review the DQB sticky notes: are learners moving from calculation to interpretation, or are they still describing procedures? The driving question demands decision-making language ('the officer should use median BECAUSE…'), not just mathematical language ('the median is 10.5'). (4) Which learners have not yet spoken aloud in cold-calls across Lessons 1–5? Ensure those learners are deliberately included in Lesson 6's cold-call rotation — every voice in the room is a data point this class cannot afford to ignore.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners physically lined up as data points to locate the middle value of an 11-value Nakuru harvest subset, and compared mean vs median for a Kisumu KCPE dataset with and without an outlier score.
What did I learn?	The median is the middle value of an ordered dataset, found at position (n+1)/2 for odd n or by averaging the two middle values for even n; unlike the mean, the median is not pulled away by extreme values (outliers).
How does this explain the phenomenon?	For the Nakuru maize harvest data, the median (approx. 10.5 bags) is lower than the mean (14.6 bags) because a few farmers with very large harvests pull the mean upward — making the median a fairer representation of what a typical Nakuru smallholder farmer actually harvests, and therefore a better number for the county government to use when allocating subsidised fertiliser.

LESSON 6 (40 minutes): Drawing the Journey — Line Graphs

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students will construct and interpret line graphs using real Kenyan data sets, identifying trends, peaks, and troughs to communicate patterns that support agricultural and economic decision-making.
Knowledge	Students know that a line graph displays how a quantity changes over time by connecting plotted points with straight line segments; they know the terms trend, peak, and trough; they know that the horizontal axis typically shows time intervals and the vertical axis shows the measured quantity.
Skills	Students can choose an appropriate scale for each axis, accurately plot data points, connect points with ruled line segments, label axes with units, and read off values and trends from a completed line graph.
Attitudes	Students appreciate that visual representations of data reveal patterns hidden in raw numbers and that choosing the right graph type is a responsibility when communicating information to decision-makers.
Key Inquiry Question	How do we collect, organize, and represent data? (KICD Key Inquiry Question 1)
Purpose in Storyline	Having computed mean, mode, and median in Lessons 3 to 5, students now gain a visual tool that reveals HOW data changes over time — a dimension none of the averages could show. The Kericho rainfall phenomenon provides the hook; monthly maize prices in Nairobi provide the practice data. This new evidence will later be layered with bar graphs and pie charts in Lessons 7 and 8 before students write their final advisory report in Lesson 9.
Safety Notes	No physical safety hazards. Remind students that graph paper edges are sharp; rulers should be handled carefully. Ensure all data values used are age-appropriate and do not identify individual students.

B. LESSON OVERVIEW

The lesson opens by revisiting the Nakuru maize harvest puzzle: the officer now asks whether the harvest changed month by month — a question that mean, mode, and median cannot answer alone. Students predict what a visual record of change over time might look like. They then observe a short teacher-led demonstration using Kericho monthly rainfall data plotted step-by-step on the board, watch a brief embedded video clip showing real KNBS maize price data being graphed, and engage in a structured model-building activity where pairs construct their own line graphs of monthly maize retail prices in Nairobi (January to December). The lesson closes with a DQB update: students post new evidence about how line graphs add a temporal story to the statistical toolkit already built.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Video: Least-Squares Regression]	Students recall from Lesson 5 that mean, mode, and median each gave ONE number to represent the Nakuru harvest data. The teacher displays the original 20-	Display the original Nakuru chalkboard data. Ask: 'Our mean was about 17 bags, our median was 16 bags. But did farmers harvest more in March than in	Elicit-before-instruct: surfacing prior conceptions about representing change before introducing the line graph. Anchoring phenomenon	Teacher reads student written predictions during circulation to identify those who already sketch something like a graph (extend), those who describe a table

Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	value chalkboard list from the anchoring phenomenon and asks: 'The KALRO officer now wants to know whether maize yields in Nakuru went UP or DOWN over the past 12 months. Can our three averages answer that question?' Students think individually for 60 seconds, then share with a shoulder partner. After 30 seconds of pair talk, three or four pairs share aloud. The teacher records responses on one side of the board under the heading 'What our averages CAN tell us' and 'What they CANNOT tell us.' Students are then shown a single question written on a card: 'If you wanted to show a friend how your school attendance changed every week this term, how would you do it — in words, a table, or a picture?' Students write one sentence in their exercise books predicting what a 'picture of change over time' might look like before any instruction is given.	October? Can a single number tell us that?' WAIT 45 SECONDS for silent thinking. Cold-call two students by name. Accept all answers without evaluation — note them on the board. Then say: 'Today we are going to find a tool that shows the JOURNEY of data, not just a snapshot.' Show the attendance question card. Say: 'Write your prediction. No wrong answers — I want your honest first thought.' WAIT 60 SECONDS. Circulate and read two or three student responses silently to gauge prior knowledge. Do not correct yet.	reconnect: linking the new tool explicitly to the unresolved question from the chalkboard scene.	(bridge), and those who say they would use words only (support). Mental note of three students in each category to monitor through the lesson.
Observe Phase 📹 **VIDEO:** Video: Least-Squares Regression Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a large set of axes on the board and says: 'Let us look at Kericho — one of Kenya's most important tea and rainfall zones.' A table of Kericho mean monthly rainfall (mm) is projected or written on the board: Jan 86, Feb 97, Mar 151, Apr 218, May 196, Jun 89, Jul 63, Aug 71, Sep 97, Oct 147, Nov 143, Dec 104. The teacher models the construction process out loud, making each decision visible: choosing the horizontal axis for months, choosing the vertical axis scale (0 to 250 mm in steps of 50), labeling both axes with units, plotting each point with a small cross, then joining successive points with a ruler. Students watch	Before drawing, say: 'Watch my decisions — I am going to think aloud so you hear WHY I choose each thing.' As you place the horizontal axis label, say: 'Months go on the horizontal axis because time always goes left to right on a line graph.' As you choose scale, say: 'My highest value is 218. If I use steps of 50, how many grid lines do I need? Count with me.' WAIT 10 SECONDS for choral count. After plotting all points, ask: 'Which month had the highest rainfall? How can you tell without reading the number?' WAIT 30 SECONDS. Take two responses. Introduce the word PEAK. Ask: 'What is the opposite of a peak in data language?' WAIT 20	Think-aloud modelling: externalising expert decision-making during graph construction. Choral prediction during video: low-stakes participation that keeps all learners actively predicting rather than passively watching. New vocabulary (peak, trough, trend) introduced in context of a real Kenyan dataset.	Thumbs-up or thumbs-down during video second viewing gives instant whole-class read of whether students can read trends from a partially completed graph. Teacher notes any students consistently incorrect and plans to seat them next to a peer during the model-building phase.

	and annotate a printed or copied axes template in their books simultaneously, mirroring each step. Once the graph is complete, the teacher circles the highest point (April, 218 mm) and labels it 'peak' and circles the lowest point (July, 63 mm) and labels it 'trough.' The teacher then traces the overall direction and introduces the word 'trend.' Students are next shown a 3-minute video clip (Kenya Meteorological Department style) in which a presenter reads aloud monthly maize retail prices from a KNBS bulletin and a simple line graph builds on screen point by point. Students watch once, then the teacher pauses after each plotted point in a second viewing and asks the class to call out whether the next point will be higher or lower — a choral prediction activity.	SECONDS. Accept TROUGH. During the video second viewing, pause before each new point and say: 'Thumbs up if you think the next point is higher, thumbs down if lower.' Scan the room for consensus before revealing the point.		
Explain Phase 📹 **VIDEO:** Video: Least-Squares Regression Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a structured class discussion anchored by three questions written on the board. Question 1: 'What does a line graph show that a frequency table cannot?' Question 2: 'Why do we JOIN the points with lines rather than leave them as separate dots?' Question 3: 'If the Nakuru KALRO officer had a line graph of monthly maize prices, what decision could she make that she could NOT make from the mean alone?' Students discuss Question 1 in pairs for 90 seconds, then share. The teacher scribes key ideas: 'change over time,' 'trend,' 'going up or going down,' 'season.' For Question 2, the teacher probes: 'Is there a value between January and February?' leading students to articulate that joining points is appropriate when the	Post the three questions on the board before discussion begins. For Question 1, say: 'Do not answer yet — talk to your partner first. I want to hear YOUR ideas before I say anything.' WAIT 90 SECONDS of pair talk. Cold-call one pair from the support group identified in the Predict phase — affirm their contribution. For Question 2, say: 'Here is a harder question. Between 1st January and 1st February, did rain stop falling completely?' WAIT 20 SECONDS. 'So is rainfall a continuous thing?' Guide students toward the concept. For Question 3, say: 'I want you to write — do not say anything yet — write one sentence starting with the words: The KALRO officer could decide...' WAIT 2 MINUTES. Read two student responses aloud (with	Structured academic controversy format for the three questions — pair discussion before whole-class share ensures every student has rehearsed an answer. Written sentence stem for Question 3 gives scaffolded access to higher-order thinking. Checklist provides procedural anchor students can self-monitor against during model building.	Teacher collects the written sentence for Question 3 as an exit check of conceptual understanding before model building begins. Sentences that mention 'time' and 'decision' are on track; those that restate the definition of mean need a brief one-to-one clarification during the model-building transition (30 seconds each).

	variable changes continuously over time. For Question 3, students work individually for 2 minutes to write one sentence answering the question, then four students share. The teacher records a class consensus statement on the board: 'A line graph shows the journey of data over time — it answers WHEN as well as HOW MUCH.' Students copy this statement into their books as the key conceptual takeaway. The teacher then explicitly states the construction rules as a numbered checklist students copy: (1) Label horizontal axis with time intervals and units. (2) Label vertical axis with the measured quantity and units. (3) Choose a scale that fits all values with room above the highest point. (4) Plot each point accurately using a cross or dot. (5) Join consecutive points with a straight ruled line. (6) Give the graph a title.	permission), then invite two more students to share theirs. Consolidate by writing the consensus statement. When delivering the checklist, say: 'These six rules are non-negotiable in any exam or report. Number them exactly as I write them.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Least-Squares Regression] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Equivalent Fractions: Number Line (PLIX)] Source: CK-12	The teacher directs students to the class Driving Question Board, which shows the original anchoring question from Lesson 1: 'Which single number best represents all our farmers — and how do we even begin to make sense of all this data?' Students are reminded of the sticky-note evidence already posted: 'Frequency tables organise data' (Lesson 1 and 2), 'Mean gives the balance point' (Lesson 3), 'Mode shows the most common value' (Lesson 4), 'Median resists outliers' (Lesson 5). Each student receives one sticky note and writes a new piece of evidence in the format: 'A line graph can help the KALRO officer because...' completing the sentence. Students	Point to each existing DQB sticky-note cluster and do a 20-second lightning recap: 'Lesson 3 gave us the mean — what was it for the Nakuru data? Shout it out.' WAIT 5 SECONDS. 'Lesson 5 gave us the median — what was it?' WAIT 5 SECONDS. 'Now Lesson 6 gives us something those numbers could not give us. Write your evidence note.' WAIT 2 MINUTES for writing. As students post notes, read two silently. Then say: 'I am going to read three of these. After each one, give a thumbs-up if you agree it is new evidence we did not have before today.' Read three. After posting the class consensus note, ask: 'What are we still missing? What question do we	Evidence accumulation on the DQB makes the learning progression visible and honours each lesson as a building block. The sentence stem ensures students frame their new learning as evidence rather than isolated fact. The open-question close maintains productive tension and curiosity for the next lessons.	Quality of DQB sticky notes is a written formative check: notes that name a specific decision the graph enables show transfer; notes that only describe the graph mechanically show surface understanding. Teacher sorts the three categories mentally and adjusts the opening of Lesson 7 accordingly.

🔍 Search ARES for similar readings	post their notes in a new column labelled 'Lesson 6 — Visual Evidence.' The teacher reads three notes aloud and invites brief comments. The class agrees on one new DQB insight to write in a shared colour: 'A line graph reveals TRENDS over time that a single average cannot show — it answers the question of HOW the harvest is changing, not just what it usually is.' The teacher connects this to the driving question: 'We now have numbers (mean, median, mode) AND pictures of change (line graphs). Are we closer to helping the Nakuru county government? What are we still missing?'	still have on the board that we have not answered?' WAIT 30 SECONDS. Accept two responses and leave them as open questions for Lessons 7 and 8.		
Model Building Phase 📹 **VIDEO:** Video: Least-Squares Regression Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Equivalent Fractions: Number Line (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students work in pairs. Each pair receives a printed data card showing monthly maize retail prices (Ksh per 2-kg packet) in Nairobi for one calendar year, sourced from a fictional KNBS bulletin: Jan 78, Feb 75, Mar 72, Apr 80, Sep 88, Oct 91, Nov 95, Dec 98. Students also have graph paper or a printed axes template. Using the six-rule checklist copied in the Explain phase, pairs construct a complete line graph. They must: (1) decide and write the scale for the vertical axis, (2) label both axes with correct units, (3) plot all 12 points, (4) join points with a ruler, (5) write a title. When the graph is complete, each pair answers three interpretation questions written on the board: (A) In which month was maize most expensive? (B) Describe the overall trend from January to December. (C) If you were the Nakuru County food security officer, what would you tell farmers to do with their harvest based on	Circulate during the construction phase. At the 5-minute mark, stop the class briefly and ask: 'Raise your hand if you have chosen your vertical axis scale.' For those who have not, ask: 'What is your highest value? What is a sensible step that gets you there without too many lines?' WAIT 15 SECONDS then move on. At the 10-minute mark, check that points are being plotted with crosses and joined with a ruler — correct any freehand curves by saying: 'Line graphs use STRAIGHT lines between points — we are not guessing the curve, we are connecting what we measured.' For Question C, circulate and prompt with: 'Think like the KALRO officer from our video. What does the high price in November and December tell a farmer about WHEN to sell?' WAIT 20 SECONDS per pair before giving the prompt. For the extension pairs, say: 'When two lines are on one graph, you MUST use	Pair construction with the checklist as a self-monitoring tool builds procedural fluency while the interpretation questions (especially Question C) require transfer back to the anchoring phenomenon context. The extension task introduces the concept of comparative line graphs, previewing Lesson 7 bar graph comparisons. Peer display and verbal explanation at the close is a low-stakes presentation that consolidates understanding through teaching.	Teacher uses a clipboard checklist with three criteria per pair: (1) scale chosen correctly and labeled with units — yes or no; (2) all 12 points plotted accurately — yes or no; (3) Question C answer references a real decision, not just a description of the graph — yes or no. Pairs with two or more 'no' marks receive targeted feedback before the lesson ends. Question C answers are collected as written evidence for the teacher portfolio.

	this graph? Question C requires a written sentence of at least two lines. Pairs that finish early receive an extension card: 'Add a second line to your graph showing prices in Kisumu (data provided) and describe the difference between the two cities.'	different colours and a KEY — can you add that?' Close the activity by selecting two pairs with visibly different graph layouts to display their work on the board and explain one feature each to the class.		

D. TEACHER REFLECTION

After the lesson, the teacher considers: Did students distinguish between the LINE GRAPH as a tool for continuous or time-ordered data and the FREQUENCY TABLE as a tool for categorical counts? Were students able to articulate in Question C WHY the graph matters for a real decision — or did they stay at the level of describing features? Did the DQB sticky notes show genuine new evidence, or were students repeating prior lessons? Identify the three pairs who found scale selection most difficult — in Lesson 7, seat them near peers who chose scales fluently so they can observe the reasoning process again during bar graph construction. Note whether the extension task was accessible to the fastest pairs — if not, prepare a more demanding extension for Lesson 7 involving a grouped frequency distribution graph.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that monthly maize retail prices in Nairobi changed every month throughout the year, with prices lowest in the middle of the year and highest toward the end of the year, forming a visible pattern when the data points were connected.
What did I learn?	We learned that a line graph is constructed by plotting data points on labeled axes and joining consecutive points with straight ruled lines; it shows trends, peaks, and troughs in data that changes over time, providing information that mean, mode, and median alone cannot reveal.
How does this explain the phenomenon?	We explained that the KALRO officer in the Nakuru phenomenon could use a line graph of monthly maize prices to advise farmers on the best time to sell their harvest — something no single average could communicate — giving the county government richer evidence for its fertilizer allocation and food security decisions.

LESSON 7 (80 minutes): Comparing with Bars: Constructing and Interpreting Bar Graphs	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students construct and interpret bar graphs from frequency distribution tables to compare maize harvest data across multiple Rift Valley counties, developing the ability to communicate data stories visually for real-world decision-making.
Knowledge	Students will know that a bar graph represents categorical or grouped data using rectangular bars whose heights (or lengths) correspond to frequencies or values, enabling visual comparison across categories.
Skills	Students will be able to: (1) construct an accurate bar graph from a frequency distribution table using a suitable scale, labelled axes, and a title; (2) read and interpret bar graphs to make comparisons between categories; (3) identify the category with the highest and lowest frequency from a bar graph.
Attitudes	Students appreciate the power of visual data representation in communicating information clearly and efficiently to support community decision-making, and demonstrate accuracy and neatness in graph construction.
Key Inquiry Question	How do we collect, organize, and represent data?
Purpose in Storyline	In Lessons 1–2 students organized the Nakuru maize harvest data into frequency distribution tables. In Lessons 3–6 they explored measures of central tendency (mean, median, mode). Now in Lesson 7, students translate those tables into bar graphs so that the KALRO officer — and the county government — can visually compare harvest yields across five Rift Valley counties at a glance, moving one step closer to answering: which number AND which visual best represents farmers' harvests for policy decisions?
Safety Notes	No physical safety hazards. Remind students to use rulers carefully when drawing axes and bars. Ensure graph paper or squared exercise-book pages are used to promote accuracy.

B. LESSON OVERVIEW

Lesson 7 opens by reconnecting students to the Nakuru Maize Harvest Mystery. The teacher shows a brief video clip of a KALRO data officer presenting a raw frequency table of maize harvests (in 90-kg bags) from five Rift Valley counties — Nakuru, Uasin Gishu, Trans-Nzoia, Nandi, and Kericho — to county government officials who look confused by the numbers alone. Students are challenged: 'Can you turn this table into a picture that even a busy county governor can understand in five seconds?' Students first predict what a bar graph might look like, then watch a focused instructional video on constructing bar graphs. They then construct their own bar graphs from the frequency distribution tables built in earlier lessons, compare harvest patterns across the five counties, and discuss how bar graphs make comparisons immediate and powerful. The lesson concludes with students updating the Driving Question Board and revising their cumulative data models.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [WATCH: The Agricultural	Students are presented with a frequency distribution table showing total maize harvest categories (e.g., 1–5 bags, 6–10	Display the frequency distribution table on the board (pre-drawn or projected). Say verbatim: 'Before we learn anything new today, I	Elicit-before-Instruction (Predict-First Sketching): By asking students to sketch a bar graph before formal instruction, the	Observable indicator: Student sketches show some attempt to use rectangular bars and relate bar height/length to frequency

Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	bags, 11–15 bags, 16–20 bags, 21–25 bags, 26–31 bags) and the number of farmers in each category across five Rift Valley counties. Without any instruction on bar graphs, students are asked: 'If you had to draw a picture — using only rectangles — that shows which harvest category was most common in Nakuru County, what would it look like? Sketch it in your exercise book and write one sentence explaining your thinking.' Students work individually for 5 minutes, then share with a shoulder partner for 2 minutes.	want you to trust your instincts. Look at this table carefully. In your exercise book, sketch a picture using only rectangles that shows the harvest data. You have 5 minutes — start now.' Set a visible timer. After 5 minutes, say: 'Share your sketch with your partner. Explain why you drew it that way. You have 2 minutes.' After partner share, cold-call 3 students to describe their sketches to the class. Ask each: 'What did you put on the horizontal line? What did you put on the vertical line? Why?' Allow 10–12 seconds of WAIT TIME after each question before accepting answers. Record key vocabulary students use naturally (e.g., 'height,' 'columns,' 'number of farmers') on the board without correcting — these become anchors for the video phase.	teacher surfaces prior knowledge and intuitive visual-spatial reasoning. This creates a cognitive hook — students will actively compare their mental model against the correct technique during the video, deepening retention.	values, even if imprecise. Listen for students using the words 'height,' 'more,' 'less,' 'category,' or 'number of farmers.' Flag any students who draw pie-like or line-like sketches for targeted support during the Observe phase.
Observe Phase 📹 **VIDEO:** WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students watch a 12–15 minute instructional video focused on constructing bar graphs. The video uses the exact Nakuru County maize harvest data from the anchoring phenomenon, walking through: (1) drawing and labelling the horizontal axis (harvest categories in 90-kg bags) and vertical axis (number of farmers / frequency); (2) choosing an appropriate scale for the vertical axis; (3) drawing bars of uniform width with gaps between them; (4) adding a title, axis labels, and units. The video then shows a completed comparative bar graph for all five Rift Valley counties side by side, and a KALRO officer saying: 'Now I can see in one glance that Trans-Nzoia farmers consistently harvested more bags than Nandi farmers — I don't need to read every number.' Students	Before playing the video, say verbatim: 'As you watch, keep your sketch in front of you. Notice: where were you right? Where do you need to adjust? Complete the note-frame on page ___ of your activity sheet — pause your thinking and write every time the video pauses.' Pause the video at the moment the scale is chosen and ask the class: 'Why do you think the video chose a scale of 0 to 10 going up by 2s rather than going up by 1s? Discuss with your partner for 1 minute.' Cold-call 2 students after WAIT TIME of 12 seconds. Resume video. After the video ends, say: 'Look at your original sketch. Hold it up. Give yourself a thumbs up for every feature your sketch already had correctly.' Allow 30 seconds of quiet reflection. Then ask: 'What is ONE thing the video showed that	Anchored Video Note-Taking with Sketch Comparison: Students watch the video with their own prediction sketch beside them, actively comparing what they intuited with what is being taught. The structured note-frame focuses attention on the four key components of a bar graph (axes, scale, bar height, gaps). The mid-video pause builds metacognitive awareness about scale selection — a common source of error.	Observable indicator: Completed note-frames contain all four blanks filled in with accurate language (horizontal axis = harvest categories; vertical axis = number of farmers; bar height = frequency; gaps = separate categories). During the mid-video pause, listen for students articulating that larger scales avoid clutter and make the graph readable. Flag students with blank or incomplete note-frames for pair support during Phase 3.

	take structured notes using a provided note-frame: 'The horizontal axis shows ___, the vertical axis shows ___, each bar's height tells us ___, the gap between bars shows ___.'	surprised you or that you did not include in your sketch?' Cold-call 4 students. Record their responses on the board under the heading 'What the bars tell us.'		
Explain Phase 📹 **VIDEO:** WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students now construct their own bar graphs. Each group of 4 receives a printed or board-copied frequency distribution table showing maize harvest data (in 90-kg bags) for TWO assigned Rift Valley counties (pairs: Nakuru & Uasin Gishu; Trans-Nzoia & Nandi; Kericho & Nakuru — cycling across groups). On graph paper, students: (1) construct one bar graph per county; (2) write three comparison sentences of the form 'In [County A], the most common harvest category was ___ bags, while in [County B] it was ___ bags. This means ___.' (3) answer the question: 'If the county government can only send extra fertilizer to the county whose farmers harvested the LEAST, which county should they choose — and how does your bar graph prove it?' Groups then do a gallery walk (5 minutes) to view other groups' graphs, leaving one sticky-note question or comment on each poster.	Distribute graph paper and the county data tables. Say verbatim: 'You are now the KALRO data officer. Your job is to draw bar graphs that help the county government decide where to send subsidized fertilizer. Your graphs must be clear enough for a governor who has only 10 seconds to look at them. You have 20 minutes.' Circulate and use targeted prompts — do NOT correct errors immediately: Ask: 'What does the height of this bar tell the governor?' (WAIT 10 seconds). 'Is your scale the same for both graphs? Why does that matter for comparison?' (WAIT 12 seconds). 'Which bar jumps out first? Is that the story you want to tell?' After 20 minutes, initiate gallery walk: 'Stand up, take one sticky note, and visit the group to your left. Read their graphs. Leave one question or one 'I notice…' comment. You have 5 minutes.' After gallery walk, cold-call 3 groups to share: 'What was the most interesting thing you saw on another group's graph?' Allow 12 seconds WAIT TIME per response.	Construct-to-Communicate (Graph as Argument): Students are not just drawing graphs for accuracy — they are constructing a visual argument to answer the driving question. By framing the task as 'convince the governor,' students engage with the graph as a communication tool, not merely a mathematical exercise. The comparison sentences force explicit verbal articulation of what the visual shows, bridging graphical and linguistic reasoning. The gallery walk introduces peer critique and reinforces that multiple valid representations can tell the same data story.	Observable indicators: (1) Bar graphs have labelled axes (harvest category and number of farmers), a title, uniform bar widths, and gaps between bars. (2) Scale is appropriate — bars do not overflow the page and differences between bars are visible. (3) Comparison sentences correctly identify the modal class and make a valid inference. (4) The fertilizer recommendation is justified by pointing to specific bars, not just stating a county name. Collect one graph per group at the end for written feedback.
Driving Question Board (DQB) Creation 📹 **VIDEO:** WATCH: The Agricultural Revolution Source: Khan	The class gathers around the Driving Question Board (DQB). The teacher reads aloud the original driving question: 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' Students are given 3 minutes to individually	Point to the DQB and say verbatim: 'Look at everything we have added to this board since Lesson 1. We started with a chaotic list of 20 numbers on a chalkboard in Nakuru County. What can we do with that data now that the KALRO officer could not do just by looking at the raw list?'	Cumulative Storyline Mapping on the DQB: The DQB functions as a public, visible record of the class's growing understanding. By physically drawing connector lines between lesson tools, the teacher makes the coherence of the evidence-gathering sequence explicit. Students see that bar	Observable indicators: Sticky notes under 'Evidence We Now Have' correctly name bar graph construction and/or visual comparison as new capabilities. New questions show genuine curiosity and conceptual reach (e.g., questions about histograms, pie charts, or comparing more

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	write on a sticky note: (1) ONE thing they can now do with data that they could NOT do in Lesson 1 (representing data as a bar graph); and (2) ONE new question that today's lesson raised for them (e.g., 'What if we want to compare more than two counties on the same graph?' or 'What is the difference between a bar graph and a histogram?'). Students place their sticky notes on the DQB under the column 'Evidence We Now Have' and 'New Questions.' The teacher reads 4–5 sticky notes aloud and draws a thread on the DQB connecting today's bar graph skill to the frequency tables from Lessons 1–2 and the mean/median/mode from Lessons 3–6, saying: 'Our tools are building on each other — tables helped us count, averages helped us summarize, and now bar graphs help us COMPARE and COMMUNICATE.'	Allow 10 seconds of WAIT TIME. Cold-call 3 students. Then say: 'Write your sticky notes — one thing you CAN now do, one NEW question you still have. 3 minutes.' After students place notes, read 4–5 aloud and physically draw connector lines on the DQB between 'Frequency Table (L1–2),' 'Mean/Median/Mode (L3–6),' and 'Bar Graph (L7).' Say: 'Two more lessons to go. What do you think we still need to learn to fully answer our driving question?' Cold-call 2 students. Record their predictions on the DQB under 'Coming Up.'	graphs are not an isolated skill but a natural extension of the tables and averages they have already built — each tool answers a different aspect of the driving question.	datasets). Flag students with sticky notes that only name the procedure ('I can draw bars') without linking to meaning for follow-up discussion in Lesson 8.
Model Building Phase 📹 **VIDEO:** WATCH: The Agricultural Revolution Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Circle Graphs to Make Bar Graphs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students return to their cumulative 'Data Story Model' — a living document/poster they have been building since Lesson 1 that represents the KALRO officer's journey from raw data to insight. In earlier lessons, students added: a raw data list, a tally and frequency table, mean/median/mode calculations, and written interpretations. Today, students add a new panel to their model: a neatly constructed bar graph of the Nakuru County maize harvest data (using the frequency table from Lesson 1), a caption reading 'A bar graph shows COMPARISON at a glance,' and an arrow pointing from the bar graph back to the frequency table with the label 'Built from this table.' Students also	Say verbatim: 'Open your Data Story Model. Today we add the bar graph panel — it connects directly back to the frequency table you drew in Lesson 1. Draw your Nakuru County bar graph neatly, add the caption and the arrow. You have 10 minutes.' Circulate and prompt: 'Can someone looking at your model for the first time follow the journey from raw data to bar graph? What would help them?' (WAIT 10 seconds). After 10 minutes: 'Now find the panel where you wrote about mean, median, and mode. Look at your bar graph. Which measure of central tendency — mean, mode, or median — do you think the KALRO officer should report to the county governor, now that you can	Cumulative Model Revision (Living Poster): The Data Story Model is a constructivist anchor — it makes visible the accumulation of understanding across the unit. Adding the bar graph panel and drawing a physical arrow back to the frequency table reinforces the idea that representations are connected and build on each other. The sentence connecting the bar graph's tallest bar to the mode or mean forces integration of Lessons 3–7, developing the cross-lesson coherence that is central to the driving question.	Observable indicators: (1) Bar graph panel is accurately constructed with labelled axes, title, and appropriate scale. (2) The caption and arrow correctly link the graph to the frequency table. (3) The revised 'Best single number' sentence makes a defensible claim connecting a specific measure of central tendency to a visible feature of the bar graph (e.g., 'The mode is easiest to read from the bar graph because it is the tallest bar'). Collect 5 randomly selected models for written feedback before Lesson 8.

	revise the 'Best single number' panel from Lessons 3–6: 'Now that we have a bar graph, does the mean or the mode connect more clearly to the tallest bar? Write one sentence.' Students share their revised models with a new partner (not their usual group) for 3 minutes of peer feedback.	SEE the data? Write one sentence and be ready to defend it.' Allow 5 minutes. Cold-call 4 students to share their sentence; for each, ask the class: 'Do you agree? Thumbs up or thumbs sideways.' After peer share, close by saying: 'In Lessons 8 and 9, we will add two more tools to this model. By the end, your Data Story Model will tell the complete story of how we turned a chaotic harvest list into a decision that could help every farmer in the Rift Valley.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully transfer the frequency distribution tables from Lessons 1–2 into accurately constructed bar graphs, or were there persistent errors in scale selection or axis labelling that need to be addressed at the start of Lesson 8? (2) During the gallery walk, did students' comparison sentences demonstrate genuine inferential thinking ('This means Nandi farmers may need more support') or did they remain purely descriptive ('Nandi has a shorter bar')? What scaffolding could deepen inference-making in Lesson 8? (3) Were students able to articulate a connection between the bar graph's tallest bar and the mode identified in earlier lessons? If not, plan a 5-minute bridge activity at the start of Lesson 8 using a simple side-by-side display of the frequency table, the mode calculation, and the bar graph. (4) Review the sticky notes from the DQB: do students' new questions point naturally toward the content of Lessons 8 and 9 (e.g., histograms, line graphs, pie charts)? If so, use 2–3 of these student-generated questions as the opening hook for Lesson 8. (5) Check the 5 collected Data Story Models: is the arrow connecting the bar graph back to the frequency table present and labelled correctly? This connection is the conceptual heart of the lesson — students who omit it may still be treating bar graphs as a standalone skill rather than part of a connected data-representation toolkit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students watched a KALRO officer present a frequency distribution table of maize harvest data from five Rift Valley counties (Nakuru, Uasin Gishu, Trans-Nzoia, Nandi, Kericho) to county government officials and observed how raw table data is difficult to interpret quickly. They also watched an instructional video demonstrating how to construct a bar graph from a frequency distribution table using the Nakuru County maize harvest data from the anchoring phenomenon.
What did I learn?	Students learned that a bar graph uses rectangular bars whose heights represent the frequency of each category, requires labelled horizontal and vertical axes (categories and frequency respectively), a suitable scale, uniform bar widths, gaps between bars, and a descriptive title. They also learned that bar graphs enable immediate visual comparison between categories — making it possible to identify the most and least common harvest ranges at a glance — and that the tallest bar corresponds to the modal class identified in earlier lessons.
How does this explain the phenomenon?	Students explained how bar graphs constructed from the Rift Valley counties' frequency distribution tables allow the KALRO officer to visually compare harvest yields across counties and identify which county's farmers harvested the least — supporting the county government's decision about where to allocate subsidized fertilizer. They also explained the connection between the mode (most frequent value, Lessons 3–6) and the tallest bar on the graph, demonstrating that different representations of the same data reveal

	the same underlying story in different ways.

LESSON 8 (80 minutes): Slicing the Pie: Constructing Pie Charts from Community Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students construct accurate pie charts by hand using percentage-to-angle conversion, applying this skill to the class dataset collected in Lesson 4 to produce a visual representation that supports decision-making for Nakuru County.
Knowledge	Students will know that a pie chart represents parts of a whole as sectors of a circle, that each sector's angle is calculated as (frequency ÷ total) × 360°, and that the sum of all sector angles must equal 360°. Students will also know that pie charts are most effective when comparing proportions across a small number of categories.
Skills	Students will be able to convert raw frequencies to percentages, calculate sector angles, use a protractor and compass to construct a pie chart accurately by hand, and label each sector with its category and percentage. Students will also be able to interpret a completed pie chart to identify the dominant category in a dataset.
Attitudes	Students will appreciate the power of visual data representation in communicating information clearly to non-specialist audiences such as county government officials and farmers in Nakuru County. Students will demonstrate precision, patience, and pride in producing accurate, well-presented mathematical work.
Key Inquiry Question	How do we collect, organize, and represent data?
Purpose in Storyline	In Lesson 4 students collected class data on a community variable. By Lesson 7 they had organized it into frequency tables and calculated measures of central tendency. Now in Lesson 8 they 'slice the pie' — converting that same dataset into a pie chart. This gives the KALRO officer and county government a second powerful visualization tool (alongside the bar/line graphs from earlier lessons) that highlights proportions at a glance, advancing the driving question by showing decision-makers not just how many farmers harvested each amount but what fraction of the farming community each group represents.
Safety Notes	Students will use compass points and protractors. Remind learners to handle compass points with care and to keep them flat on the desk when not in use. No other safety concerns for this lesson.

B. LESSON OVERVIEW

Students open the lesson by revisiting the Nakuru County maize harvest driving question and recalling the class dataset collected in Lesson 4. The teacher uses the Kenya National Energy Mix (solar, hydro, geothermal, wind, fossil fuels) as a concrete, relatable supporting phenomenon to model the full pie-chart construction process step by step — from percentage calculation to angle calculation to drawing with compass and protractor. Students then replicate the process independently using the class maize-harvest dataset, producing a hand-constructed pie chart they annotate and compare with a partner. The lesson closes with a DQB update where students debate whether a pie chart or a bar chart better serves the KALRO officer's reporting needs, deepening their understanding that different representations reveal different aspects of the same data.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown a blank circle	Display the blank circle and the	Prediction with Justification — by	Observable indicator: Students

📹 **VIDEO:** [Video: Energy Skate Park: Basics](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Interpretations of Circle Graphs: Northwest Region Population Pie (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	drawn on the board and the class maize-harvest frequency table summary from Lesson 4 (e.g., Harvest Range: 1–10 bags, 11–20 bags, 21–31 bags, with frequencies). Working individually for 3 minutes, each student sketches — by eye, without any calculation — how they would 'slice' the circle to show the three groups of farmers. They write a one-sentence justification: 'I gave the largest slice to ___ because ___.' Students then share predictions with a shoulder partner.	grouped frequency table on the board (or projected screen). Say: 'Before we learn any formula, I want you to use your mathematical intuition. Look at this table — look at how many farmers fall into each harvest group. Now sketch how you would divide this circle to represent those groups. You have 3 minutes — go.' Set a visible timer. After 3 minutes, cold-call 3 students (choose one from each estimated performance level): 'Amina, show us your circle and explain your thinking.' After each response, ask the class: 'Does anyone agree with Amina? Raise your hand. Does anyone disagree? Why?' Allow 10 seconds of WAIT TIME after each question before accepting responses. Record 2–3 different student predictions on the board using quick sketches. Say: 'We will come back to your predictions at the end of the lesson to see how close your intuition was to the mathematically accurate answer.'	committing to a sketch and a written reason before instruction, students activate proportional reasoning and create a cognitive anchor. Comparing predictions to the mathematically derived result later in the lesson deepens conceptual understanding of why the calculation matters.	produce a labelled sketch of a divided circle with a written justification sentence. Listen for whether students reference relative frequency ('this group has the most farmers, so it gets the biggest slice') rather than arbitrary guessing — this reveals prior understanding of proportionality. Note any student who divides the circle into equal thirds regardless of frequency; this is a key misconception to address in Phase 3.
Observe Phase 📹 **VIDEO:** [Video: Energy Skate Park: Basics](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Interpretations of Circle Graphs: Northwest Region Population Pie (PLIX)](#) Source: CK-12 🔍 Search ARES	The teacher presents the Kenya Energy Mix supporting phenomenon: 'Kenya generated electricity in 2023 from these sources — Geothermal: 43%, Hydro: 24%, Wind: 14%, Solar: 9%, Fossil Fuels & Others: 10% — (figures displayed on board). A KETRACO engineer needs to present this data to Parliament using a pie chart.' Students observe a two-part worked model: (A) The teacher constructs the pie chart live on the board, narrating every step aloud, while students follow along on their own blank circles using compass, ruler, and protractor. (B) Students then watch a short narrated animation (or	Before starting, distribute: compass, protractor, ruler, blank paper, coloured pencils for each student. Say: 'We are going to learn the exact method the KETRACO engineer uses. Watch me do each step once — then you will do the same step on your own circle.' STEP 1 — Frequency to Percentage: 'The percentages are already given here. But if they were raw frequencies, we would calculate: (frequency ÷ total) × 100. Everyone write that formula in your exercise book now.' Pause 10 seconds. STEP 2 — Percentage to Angle: 'Now we convert each percentage to an angle. The formula is: (percentage ÷ 100) ×	Worked Example with Concurrent Practice (I Do — We Do) — the teacher models each micro-step and students immediately replicate it on their own circle. This reduces cognitive load by chunking the multi-step procedure, allowing students to experience success at each stage before the complexity accumulates. Using the Kenya Energy Mix — a real, nationally significant dataset — grounds the procedure in authentic purpose.	Observable indicators: (1) Each student's angle sum equals or rounds to 360°. (2) Protractor is correctly aligned (zero line on radius, centre on circle's centre). (3) Sectors are labelled with category name and percentage. Circulate and use targeted questions: 'Show me where you placed the zero on your protractor.' 'What angle did you calculate for Solar?' Flag students whose angles do not sum to 360° and prompt: 'Recheck your multiplication — use (percentage ÷ 100) × 360.'

for similar readings	teacher-drawn diagram sequence on board if offline) showing the same energy data pie chart being constructed. Students complete a structured observation guide: they record the angle calculated for each energy source and check that angles sum to 360°.	360°. Let's do Geothermal together: (43 ÷ 100) × 360 = 154.8°. Round to 154.8°.' Write on board. Cold-call 2 students to calculate Hydro and Wind aloud. STEP 3 — Drawing: 'Use your compass to draw a circle of radius 5 cm. Mark the centre. Use your ruler to draw one radius line to the top (12 o'clock). Place your protractor — zero on that line, centre on the centre point. Measure 154.8°, mark it, draw the next radius.' Demonstrate slowly. Walk the room after each step; check that students' protractors are aligned correctly. Say: 'Check your partner's protractor placement before you draw the line.' STEP 4 — Label and colour each sector. STEP 5 — Verify: 'Add all your angles. What should they sum to?' WAIT 10 seconds. Accept responses. 'Yes — 360°. If yours does not, find the error before moving on.'		
Explain Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Interpretations of Circle Graphs: Northwest Region Population Pie (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students now apply the full construction process independently to the class maize-harvest dataset from Lesson 4. Each student receives (or copies from board) the grouped frequency table: e.g., 1–10 bags (8 farmers), 11–20 bags (7 farmers), 21–31 bags (5 farmers), Total: 20 farmers. Working individually, they: (1) convert each frequency to a percentage, (2) convert each percentage to a sector angle, (3) construct the pie chart with compass and protractor, (4) label each sector with harvest range and percentage, (5) write a two-sentence interpretation: 'The largest slice shows ___ because ___. This tells the KALRO officer that ___.' After individual work (20	Display the grouped frequency table on the board. Say: 'This is our class data — real numbers about a real community context. Your job is to turn this table into a pie chart that the KALRO officer could include in her report to Nakuru County government. Use every step we practised.' Allow 20 minutes of individual work. Circulate every 2–3 minutes. Ask probing questions: 'Why did you multiply by 360 and not by 100 at this step?' 'What does a 144° sector tell someone looking at this chart?' 'If one group of farmers got a much larger slice than the others, what would that suggest about maize yields in Nakuru?' If a student divides the circle into equal thirds (the Lesson 4	Transfer Task with Written Interpretation — students move from the Kenya Energy Mix model to their own community dataset, requiring genuine transfer rather than rote imitation. Writing a two-sentence interpretation forces students to connect the visual representation back to real-world meaning (the KALRO officer's decision), building statistical literacy alongside procedural skill.	Observable indicators: (1) Percentage calculations are correct: (frequency ÷ 20) × 100. (2) Sector angles are correct: (percentage ÷ 100) × 360 OR equivalently (frequency ÷ 20) × 360. (3) Angles sum to 360° (allow ±1° rounding tolerance). (4) Written interpretation explicitly references what the data means for Nakuru County farmers. Collect one written interpretation per student at the end of class as an exit artefact.

	minutes), students pair-share their completed charts and written interpretations. Pairs identify any angle discrepancies and resolve them together. Whole-class debrief: teacher cold-calls 2 pairs to share their interpretations; class evaluates whether the interpretation connects back to the driving question.	misconception flagged in Predict), kneel to eye level and say quietly: 'Look at your frequency table — do all three groups have the same number of farmers? So should they get the same slice?' After pair-share, cold-call 2 pairs for whole-class sharing. After each pair presents, ask the class: 'Does this pie chart answer part of our driving question? How? What information does it show that a frequency table alone cannot?' Allow 10 seconds WAIT TIME before taking responses.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Interpretations of Circle Graphs: Northwest Region Population Pie (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students receive two sticky notes each. On the first (green): they write one thing their pie chart reveals about the Nakuru maize harvest data that was NOT visible in the raw list or the frequency table alone. On the second (orange): they write one new question or limitation they noticed — e.g., 'Does a pie chart show us how spread out the data is within each slice?' Students post both notes on the class DQB under the column: 'Pie Charts — What We Now Know / What We Still Wonder.' The teacher facilitates a 5-minute whole-class gallery walk where students read each other's notes and place a dot sticker on any orange question they also have. The most-dotted questions are circled and flagged for Lesson 9.	Hand out sticky notes. Say: 'We have been building our answer to the driving question lesson by lesson. Today we added a new tool — the pie chart. On your green note, complete this sentence: My pie chart shows that ___ and this is useful to the KALRO officer because ___. On your orange note, write one thing a pie chart cannot tell us, or one new question you now have.' Allow 4 minutes of writing. Direct students to the DQB: 'Post your green note on the left column, orange on the right. Then do a gallery walk — read three other groups' orange notes and dot the ones that are also your question.' After gallery walk, say: 'I can see that many of you are asking whether pie charts can show us the spread or range of the data. Hold that question — it connects directly to what we will explore in Lesson 9.' Record the top two orange questions on a dedicated DQB sticky labelled 'LESSON 9 AGENDA.'	Two-Column DQB Update (Know / Wonder) — separating 'what we now know' from 'what we still wonder' makes cumulative learning visible and metacognitive. The dot-voting ritual democratically surfaces the class's most pressing remaining questions, giving students agency over the inquiry direction and creating authentic motivation for Lesson 9.	Observable indicators: Green notes correctly identify a proportion or comparison that is more visible in a pie chart than in a table (e.g., 'Most farmers — 40% — harvested only 1–10 bags'). Orange notes raise genuine statistical limitations or extensions (e.g., questions about spread, about what happens with many categories, about combining pie charts with other graphs). Flag any green note that is vague ('it shows data') for follow-up questioning.
Model Building Phase	Students return to their cumulative Data Story Model — the evolving visual explanation they have been	Say: 'Open your Data Story Model to the next blank panel — Panel 8. Sketch or paste your pie chart	Cumulative Model Revision with Annotation — requiring students to label both the power and the	Observable indicators: Panel 8 contains a correctly labelled pie chart sketch. Annotation (A)

📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Interpretations of Circle Graphs: Northwest Region Population Pie (PLIX) Source: CK-12 🔍 Search ARES for similar readings	building since Lesson 1 to answer the driving question. Today they add Panel 8: a miniature version of their completed pie chart (sketched or glued in), annotated with two labels: (A) 'What the pie chart shows the KALRO officer' and (B) 'What the pie chart cannot show.' Below the panel, students write a one-sentence 'Model Update Statement': 'A pie chart is a useful representation when ___.' Students then pair with a different partner from Phase 3 and compare Model Update Statements, negotiating a shared version that they both record. The teacher closes by projecting the Lesson 1 raw data list from the anchoring phenomenon and asking: 'Look at where we started — a chaotic list on a chalkboard. Now look at your Data Story Model. What tools have we built to turn that chaos into a story for Nakuru County?'	here. Add two labels: what it shows, and what it cannot show. Then write your Model Update Statement — start with: A pie chart is a useful representation when...' Allow 8 minutes. Circulate; prompt students who write only procedural statements ('when you know percentages') to think about the communication purpose: 'Think about who is reading the chart and what decision they need to make.' After pair-negotiation, cold-call 2 pairs: 'Read us your shared Model Update Statement.' After both, ask the class: 'Which statement is more precise — and why?' Allow 10 seconds WAIT TIME. Close by projecting the original anchoring phenomenon data list. Say: 'In Lesson 1 this list confused the KALRO officer. Look at your model. Which single representation — bar chart, frequency table, mean, median, mode, or pie chart — would you recommend she use in her report, and why?' Accept 3 responses with no judgement; say: 'We will return to this question in Lesson 9 when we bring everything together.'	limitation of the pie chart within their model prevents superficial procedural learning and builds the nuanced statistical literacy needed to genuinely answer the driving question. Comparing models in pairs introduces productive disagreement and collaborative refinement.	references proportions or dominant categories visible in the chart. Annotation (B) identifies a genuine limitation (e.g., 'cannot show exact values within each group' or 'hard to compare two datasets side by side'). Model Update Statement specifies a condition for appropriate use of a pie chart, not just a procedural description.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2, were students able to align their protractors correctly without individual correction? If more than 5 students misaligned the protractor, consider introducing a 'protractor check' peer routine in future lessons — one partner places, the other verifies before drawing. (2) In Phase 3, did students' written interpretations connect the pie chart back to the KALRO officer's decision, or did they describe the chart in purely procedural terms ('the biggest slice is 40%')? If interpretations were procedural, the next lesson should open with a modelled example of a statistical interpretation that explicitly names the audience and decision context. (3) Did the DQB orange notes reveal that students understand the limitations of pie charts (e.g., inability to show spread or exact values), or were questions superficial? If students did not raise spread/variability as a limitation, introduce this explicitly at the start of Lesson 9 before the synthesis activity. (4) Were any students unable to complete the pie chart construction in the time given? If so, consider whether the Energy Mix model in Phase 2 took too long, or whether the grouping of the frequency table needs simplification (reduce to two categories) for students who need scaffolding. (5) Did students update their Data Story Models with genuine annotations, or did they simply paste a picture? Strong model updates are a reliable indicator of conceptual understanding — use these as formative data to plan differentiated support in Lesson 9.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed the teacher construct a pie chart live using the Kenya National Energy Mix data (Geothermal 43%, Hydro 24%, Wind 14%, Solar 9%, Fossil Fuels & Others 10%), following each step from percentage to angle to sector drawing with compass and protractor. They replicated each step concurrently on their own circles.
What did I learn?	Students learned that a pie chart represents each category as a sector of a circle, that sector angles are calculated using the formula (frequency ÷ total) × 360° (or equivalently, percentage × 3.6°), and that all sector angles must sum to 360°. They also learned that pie charts are best suited for showing proportions across a small number of categories, and that they reveal dominance and relative share in a way that a frequency table or raw data list cannot.
How does this explain the phenomenon?	Students explained how the class maize-harvest data from Nakuru County can be converted into a pie chart that shows the proportion of farmers in each harvest group, and why this representation is useful for the KALRO officer and county government when making fertilizer allocation decisions. They also articulated at least one limitation of pie charts — that they do not reveal the spread or exact values within each category — and recorded this in their Data Story Models and on the DQB.

LESSON 9 (80 minutes): Reading the Full Picture	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students interpret mixed statistical graphs and pie charts drawn from Kenyan data sources, synthesize all statistical tools learned across the unit, and produce a written advisory report to the Nakuru County government recommending a fertilizer allocation strategy — thereby answering the unit's driving question with full evidential support.
Knowledge	Students know how to read and interpret bar graphs, line graphs, pie charts, and frequency distribution tables; understand when to apply mean, median, and mode; and understand why statistics matter for real-life decision-making in Kenya.
Skills	Students can extract key information from mixed graphical representations, evaluate which measure of central tendency best represents a dataset, synthesize statistical evidence into a coherent written recommendation, and critically review their own learning journey using the Driving Question Board.
Attitudes	Students appreciate the power of statistics as a tool for social good and equity, demonstrating responsibility and empathy for Kenyan smallholder farmers whose livelihoods depend on data-informed government decisions.
Key Inquiry Question	How do measures of central tendency and graphical representations together help us make sense of data in real life — and what is the best statistical story we can tell to help Nakuru County farmers?
Purpose in Storyline	This final lesson is the capstone of the unit storyline. Students have spent eight lessons gathering statistical tools — frequency tables, mean, median, mode, bar graphs, line graphs, and pie charts. Now they deploy all of these tools together to write a real advisory report to the fictional Nakuru County government. They also complete the Driving Question Board, compare their initial L1 model to their final model, and formally answer the anchoring phenomenon question: 'Which single number best represents all our farmers — and how do we even begin to make sense of all this data?'
Safety Notes	No physical safety hazards. Ensure inclusive participation — some learners may feel anxious about presenting their advisory reports. Establish a respectful peer-feedback norm before the sharing activity. Remind learners that constructive critique strengthens the report, just as checking data strengthens a statistical claim.

B. LESSON OVERVIEW

Lesson 9 is the final, synthesis lesson of the Statistics I unit. Students open by revisiting the anchoring phenomenon video clip (the KALRO officer's chaotic chalkboard of maize harvest data from Nakuru County) and their Lesson 1 initial models. They then move through a structured interpretation activity in which they read a set of three mixed Kenyan data visuals — a pie chart of fertilizer subsidy distribution across Kenyan counties, a bar graph of maize yields per county over two seasons, and a frequency distribution table of Nakuru farmers' harvests — sourced from fictional KNBS-style reports. Using all statistical tools from the unit, student groups draft a short advisory report recommending a fertilizer allocation strategy. Groups share and peer-review reports, followed by a teacher-facilitated video on the real-world importance of statistics in Kenya (KNBS context). The lesson culminates in a full Driving Question Board review: students post their final answers to both key inquiry questions, compare L1 models to final models, and formally answer the driving question as a class. The teacher leads a celebratory closure acknowledging the learning journey.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Video: Energy Skate Park: Basics] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown the original anchoring phenomenon video clip again — the KALRO officer writing 20 farmers' maize harvest values (in 90-kg bags) on the chalkboard in Nakuru County: a chaotic, unorganized list. Students are then shown their own Lesson 1 initial model (their first attempt at answering 'Which number best represents all our farmers?'). Working individually for 5 minutes, each student writes two things in their notebooks: (1) What did my Lesson 1 model say? (2) What would I say NOW — and what evidence from across the unit supports my answer? Students then share with a shoulder partner for 2 minutes before a class cold-call share-out.	Replay the original phenomenon video clip (2–3 minutes). As it plays, say: 'Watch this with fresh eyes. Think about everything we have learned since Lesson 1.' After the clip, display a projected image of a sample Lesson 1 model (hand-drawn frequency table attempt and a guess at the 'best number'). Distribute learners' own L1 models (kept in portfolios). Say: 'You wrote this in Lesson 1. Read it. Do not erase anything — those ideas are your starting point and we celebrate them.' Give 5 minutes of SILENT individual writing. Circulate and read over shoulders; note 3–4 learners with particularly interesting L1-to-now contrasts for cold-calling. After partner share, cold-call 4 learners: 'Tell us one thing your Lesson 1 model got right — and one thing you now know that your L1 model was missing.' WAIT TIME: 12 seconds after each cold-call prompt. Write key contrasting phrases on the board in two columns: 'L1 Thinking' vs. 'What We Know Now'. Do NOT correct or complete — let tension build for the lesson ahead.	Strategy: **Model Comparison Activation** — Revisiting the L1 model activates prior knowledge and creates cognitive dissonance between initial intuition and accumulated evidence. This motivates learners to use the final lesson to resolve the tension and produce a complete, evidence-based explanation.	Observable indicator: Each learner has written at least one specific statistical tool (e.g., 'I now know the median ignores outliers better than the mean') in their 'What I Know Now' response. Teacher checks notebooks during circulation. Learners who cannot name a tool are paired with a stronger peer during Phase 2.
Observe Phase 📹 **VIDEO:** [Video: Energy Skate Park: Basics] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:**	Groups of 4 receive a printed 'Nakuru County Statistical Dossier' — a two-page document containing three Kenyan data visuals: (A) A pie chart showing the percentage share of Kenya's national fertilizer subsidy allocated to 8 counties (Nakuru, Nyandarua, Uasin Gishu, Trans Nzoia, Kisumu, Meru, Kakamega, and Kiambu); (B) A comparative bar graph showing average maize yield (in 90-kg bags per acre) for	Distribute the 'Nakuru County Statistical Dossier' packs and the structured observation guide. Say: 'This dossier was prepared by a fictional KNBS analyst. Your job is to be critical readers — extract the story the data is telling about Nakuru farmers.' Assign roles within each group: Data Reader (reads values aloud), Recorder (writes findings), Challenger (asks 'are we sure?' after each claim), and Reporter (will present). Walk	Strategy: **Structured Data Interpretation with Role Differentiation** — Assigning group roles ensures every learner engages with the data visuals at their level. The 'Challenger' role specifically builds critical statistical literacy by requiring learners to interrogate every claim before it is accepted — mirroring the professional practice of a data analyst.	Observable indicator: Each group's manila paper shows a written justification (not just a circled answer) for their choice of mean, median, or mode in question (3), referencing specific values from the frequency table. Groups that only circle an answer without justification receive a teacher prompt: 'What in the table makes you say that? Point to the number that convinced you.'

Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Nakuru County across two seasons — Long Rains 2022 and Short Rains 2022 — broken down by sub-county (Nakuru North, Nakuru South, Bahati, Gilgil, Rongai, Subukia, Njoro); (C) A frequency distribution table of 20 Nakuru smallholder farmers' harvests (the same dataset from Lesson 1), now fully organized with tally marks, frequency, cumulative frequency, and a calculated mean, median, and mode. Students work in groups for 15 minutes to answer a structured observation guide: (1) Which county receives the largest fertilizer subsidy share — is this fair given yield data? (2) Which Nakuru sub-county has the greatest improvement between seasons? (3) From the frequency table, which measure of central tendency — mean, median, or mode — do you think best represents the 'typical' farmer? Why? Groups record answers on large manila paper for display.	the room with a clipboard. At the 7-minute mark, pause the class: 'I am hearing some groups debate the mean vs. median. Hold that debate — it is exactly the right question.' Cold-call 2 groups to share their answer to question (3) before the full debrief: 'Group three — which measure did you choose and what was your evidence from the table?' WAIT TIME: 15 seconds. After group work, lead a 5-minute whole-class debrief projecting each visual. Use Socratic questioning: 'If the mean harvest is 14.3 bags but the mode is 8 bags — which number would a farmer want the government to use when allocating subsidies, and why?' WAIT TIME: 12 seconds. Cold-call 3 different learners.		
Explain Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES	Each group now drafts a short (one-page) Advisory Report addressed to the fictional Nakuru County Director of Agriculture. The report must include: (1) An opening statement summarizing the typical maize harvest for Nakuru farmers, naming and justifying the chosen measure of central tendency; (2) A reference to at least one graph or pie chart from the dossier as evidence; (3) A specific fertilizer allocation recommendation (e.g., 'We recommend prioritizing sub-counties with the lowest yields') supported by data; (4) A closing statement explaining WHY organized data (not raw lists)	Before groups begin writing, display a sentence-starter scaffold on the board: 'Based on our analysis of [name the visual], we found that the [mean/median/mode] of [value] bags best represents the typical Nakuru farmer because… We therefore recommend… This was only possible because we first organized the data by…' Say: 'These sentence starters are scaffolds, not a script. Stronger groups should go beyond them.' Set a visible 15-minute timer. During writing, circulate and give targeted prompts: to groups avoiding the pie chart, say 'Your report has not yet used the	Strategy: **Evidence-Based Scientific Writing (Claim–Evidence–Reasoning)** — The advisory report structure mirrors the Claim–Evidence–Reasoning framework from NGSS Science and Engineering Practice 6 (Constructing Explanations). Learners must name a claim (the best measure), cite evidence (a specific data value or graph), and provide reasoning (why this measure fits the data distribution). This transforms statistical knowledge into communicable, decision-relevant insight.	Observable indicator: Each advisory report contains all four required components. Teacher uses a quick 4-point checklist (1 point per component) during presentations. Reports scoring 2 or below receive a written teacher prompt for revision: 'Which graph have you NOT yet used as evidence? Add one sentence that references it.' Peer feedback quality is also assessed — vague feedback ('it was good') is redirected: 'Can you name the specific statistic that made it good?'

for similar readings	made this recommendation possible — connecting directly back to the anchoring phenomenon. Groups have 15 minutes to write, then 3 minutes each to present to the class. Peer groups use a simple two-star-one-wish feedback protocol: two things the report did well statistically, one statistical tool or piece of evidence they could add to strengthen it.	subsidy data — how does county-level funding connect to your recommendation?'; to groups that only cite one measure, say 'You've used the mean. What would happen to your recommendation if you used the median instead — is it the same story?' After presentations, facilitate peer feedback. Model the two-star-one-wish protocol with the first group's report before others begin. Cold-call 2 learners from the audience after each presentation: 'What statistical evidence in that report was most convincing to you?' WAIT TIME: 10 seconds.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Video (5 minutes): Students watch a short video segment about the Kenya National Bureau of Statistics (KNBS) and how government statisticians use data to make decisions affecting millions of Kenyans — covering topics like food security, unemployment, and healthcare resource allocation. The video explicitly shows a pie chart and bar graph being used in a real KNBS press release. Part B — DQB Final Update (10 minutes): The class gathers around the physical Driving Question Board (started in Lesson 1). Each student receives two sticky notes. On the first (green), they write their final, complete answer to Key Inquiry Question 1: 'How do we collect, organize, and represent data?' On the second (blue), they write their final answer to Key Inquiry Question 2: 'How do measures of central tendency help us make sense of data in real life?' Students post their sticky notes on the DQB. The teacher facilitates a 5-minute class read-aloud of a selection of	Introduce the video: 'Before we close our board, I want you to see that everything you have learned in this unit is used every single day by real Kenyan professionals.' Play the KNBS video segment. After the video, pause and ask: 'Where in that video did you see a tool we used in this unit?' Cold-call 3 learners. WAIT TIME: 10 seconds each. Then distribute the two sticky notes. Say: 'These are not rough answers. These are your FINAL answers — write them as if you are writing a report for the KALRO officer from our very first video.' Give 5 minutes of silent writing. As students post their notes on the DQB, read 4–5 aloud (selecting a range from hesitant to confident voices). Say: 'Look at the board. In Lesson 1, our board was full of QUESTIONS. Now it is full of ANSWERS. That is what learning statistics looks like.' Point to specific sticky notes that directly answer the anchoring phenomenon question and read them aloud as a class.	Strategy: **Public Knowledge Display & Collective Synthesis** — The DQB functions as a living, collaborative knowledge artefact. Posting final answers publicly allows learners to see their own intellectual growth alongside their peers', reinforcing metacognitive awareness. The juxtaposition of Lesson 1 questions against Lesson 9 answers makes the learning journey visible and celebratory, consolidating long-term retention.	Observable indicator: Each learner's DQB sticky notes contain at least one specific statistical term (e.g., frequency table, median, bar graph, pie chart) and one connection to a real-life Kenyan context (e.g., farming, KNBS, fertilizer allocation). Sticky notes that are vague ('statistics help us understand data') receive a verbal prompt from the teacher before posting: 'Can you name one tool and one Kenyan example to make this answer complete?'

	sticky notes, highlighting the evolution from Lesson 1 questions to Lesson 9 answers — a visible record of the class's collective knowledge growth.			
Model Building Phase 📹 **VIDEO:** Video: Energy Skate Park: Basics Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Summary Statistics, Summarizing Univariate Distributions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students individually redraw or annotate their Lesson 1 model in their notebooks, transforming it into their FINAL model. The final model must show: (1) The process of organizing raw data into a frequency distribution table; (2) The calculation and comparison of mean, median, and mode for the Nakuru farmers' dataset; (3) At least one graphical representation (bar graph or pie chart) that could communicate the data to the Nakuru County government; (4) A written sentence at the bottom answering the driving question: 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' Students have 10 minutes for model revision. Three volunteers then share their final model under the document camera (or hold it up), explaining what changed from L1. The class closes with the teacher reading aloud the driving question one final time and inviting a whole-class choral response of one word that describes what statistics does for communities — students call out words and the teacher writes them on the board in a word cloud.	Distribute learners' L1 models alongside clean notebook paper. Say: 'You are not starting from scratch. You are upgrading. Use a different colour pen to add to your L1 model — so you can see exactly what changed.' Display the driving question prominently: 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' Set a 10-minute silent model-revision timer. Circulate and give specific praise: 'I see you have added the median calculation next to the mean — that is a significant upgrade.' After 10 minutes, invite 3 volunteers (aim for one lower-confidence, one mid-level, one high-confidence learner) to share under the document camera. For each, ask: 'What is ONE thing in your final model that was not in your Lesson 1 model?' WAIT TIME: 12 seconds. After the third share, read the driving question aloud slowly and deliberately. Say: 'In one word — what does statistics DO for communities like Nakuru?' Pause 10 seconds, then take rapid-fire responses from 8–10 learners and write each word on the board. Circle the word cloud and say: 'All of these words together — that is why we spent nine lessons on this unit. Well done.'	Strategy: **Annotated Model Revision as Final Explanation** — Asking learners to annotate (not replace) their L1 model makes the cognitive growth tangible and visible. The two-colour annotation technique (original vs. new knowledge) embeds metacognition directly into the artefact, allowing both learner and teacher to assess conceptual change. The word-cloud closure functions as a communal meaning-making ritual that consolidates affective and cognitive learning simultaneously.	Observable indicator: Each final model contains all four required components listed in the learner experience. The driving-question answer sentence uses at least two statistical terms and one Kenyan community reference. Teacher collects all final models as a summative portfolio artefact. During the three model-share presentations, teacher listens specifically for whether the learner can articulate the REASON a given measure or graph was chosen — not just that it was used. Learners who present without justification are asked a follow-up: 'Why did you choose that representation and not a different one?'

D. TEACHER REFLECTION

After Lesson 9, reflect on the following questions before closing the unit portfolio:

1. DRIVING QUESTION RESOLUTION — Did the majority of learners' final DQB sticky notes and advisory reports directly and specifically answer the driving question, naming both a statistical tool and a Kenyan community application? If not, which phase of the unit left the biggest conceptual gap?

2. MODEL GROWTH — When comparing L1 models to final models, were learners able to articulate WHY their understanding changed, or did they simply add more content without deeper reasoning? Identify 2–3 learners whose model growth was particularly strong and 2–3 whose growth was limited — what instructional support would you build into Lesson 3–5 of a future iteration to close that gap earlier?

3. ADVISORY REPORT QUALITY — Were learners able to synthesize across multiple data representations (pie chart + bar graph + frequency table) in one coherent argument, or did most reports rely on only one visual? Consider adding an explicit 'multi-source triangulation' mini-lesson (15 minutes) between Lessons 7 and 8 in future iterations.

4. ENGAGEMENT & EQUITY — Did lower-confidence learners participate meaningfully in the DQB posting and model-sharing phases, or did stronger voices dominate? Review your cold-call distribution log: did you reach all learners across the lesson, including girls, quieter students, and learners in the back rows?

5. STATISTICS IN REAL LIFE — Did the KNBS video land as intended — did learners make explicit connections between the video content and their own unit work? If learners seemed disconnected, consider sourcing a more locally grounded video (e.g., a KALRO or county government press briefing using actual Nakuru data) for future use.

6. UNIT COHERENCE — Looking across all 9 lessons: did the anchoring phenomenon (the Maize Harvest Mystery of Nakuru County) successfully thread through every lesson as a motivating context? Note any lesson where the phenomenon felt disconnected and revise the opening hook of that lesson for the next cohort.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students re-watched the original anchoring phenomenon clip of the KALRO officer's chaotic chalkboard of 20 Nakuru farmers' maize harvest values, read a fictional KNBS-style 'Nakuru County Statistical Dossier' containing a pie chart of fertilizer subsidy distribution across Kenyan counties, a comparative bar graph of maize yields by Nakuru sub-county across two seasons, and a fully worked frequency distribution table of the original 20-farmer dataset — and watched a short video on the Kenya National Bureau of Statistics using data to drive national decisions.
What did I learn?	Students learned how to read and extract meaning from multiple graphical representations simultaneously (pie charts, bar graphs, and frequency tables); how to evaluate which measure of central tendency — mean, median, or mode — is most appropriate for a given dataset and policy purpose; how to synthesize statistical evidence from multiple sources into a structured written advisory report using a Claim–Evidence–Reasoning framework; and how the entire statistical process — from raw data collection through organization, representation, and interpretation — is the foundation of real decisions made by Kenyan government bodies such as KNBS and county agricultural offices.
How does this explain the phenomenon?	Students explained, through their advisory reports and final annotated models, that the chaos of the original 20-farmer harvest list (the anchoring phenomenon) can only be transformed into actionable policy advice by applying the full statistical toolkit: organizing raw data into frequency tables, calculating and comparing mean, median, and mode to identify the most representative value, and selecting appropriate graphical representations to communicate findings clearly to decision-makers. They answered the driving question — 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' — by arguing that statistics is the systematic bridge between raw community experience and informed government action, as demonstrated through their fertilizer allocation recommendations to the fictional Nakuru County Director of Agriculture.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.